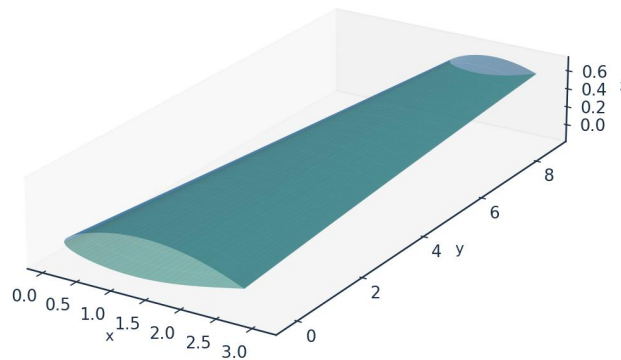


INDUSTRIAL CASE STUDY

Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

Prepared by

AKOSA SAMUEL ONYEJEKWE

1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

Key results at the cruise design point (FL360, $M=0.42$, $Re_{MAC} \approx 6.4 \times 10^6$):

Quantity	Predicted value
Upper-surface transition x_{tr}/c	0.43 (natural / Tollmien–Schlichting)
Lower-surface transition x_{tr}/c	0.58 (natural / Tollmien–Schlichting)
Mean laminar-flow extent	$\approx 52\%$ of chord
Section lift coefficient C_l	0.502
Section profile drag C_d	27.0 counts
Viscous drag reduction vs fully-turbulent	$\approx 45\%$
Climb ($Tu=0.9\%$) transition x_{tr}/c (upper)	0.05 (bypass)

Table 1. Headline predictions.

Validated against three independent, credible published transition datasets with one universal calibration set, the solver reproduces the transition-onset Reynolds number $Re_{\theta t}$ against the Rolls-Royce hot-wire measurements of ERCOFTAC case 020 and the Schubauer-Skramstad experiment. The natural and bypass branches are independent models; the separation and cross-flow branches are carried and calibrated but are not selected at any condition reported here. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, governed by the growth of the amplification envelope and predicted here by an e^N integral in Re_θ .
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

3. Problem Statement and Solution Approach

3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction C_f , momentum thickness θ , shape factor H , intermittency γ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure C_p and edge velocity U_e , with a Prandtl–Glauert correction applied to C_p and the integrated loads at the cruise Mach number. The boundary-layer closures downstream are incompressible formulations and the edge velocity passed to them is left uncorrected.
- Laminar boundary layer: Thwaites' integral method advances θ , H and Re_θ from the stagnation point along each surface.
- Unified transition kernel (the novel core): at every station the effective transition Reynolds number is the minimum across the four mechanisms, each with a calibration weight.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through γ .
- Turbulent boundary layer: Head's entrainment method with the Ludwig–Tillmann skin-friction law advances the turbulent BL to the trailing edge.

- Integration: the Squire–Young formula returns profile drag; a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file `model.equations.docx`.

4.1 Inviscid edge solution

(E02) *Vortex-panel surface velocity (Kuethe & Chow)*

$$\frac{V_{t_i}}{U_\infty} = \cos(\theta_i - \alpha) + \sum_{j=1}^N C_{t,ij} \gamma_j$$

(E03) *Kutta condition (sharp trailing edge)*

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) *Pressure coefficient (inviscid)*

$$C_p = 1 - \left(\frac{V}{U_\infty} \right)^2$$

[missing eq E04]

4.2 Laminar boundary layer (Thwaites)

(E05) *Thwaites momentum thickness (laminar)*

$$\theta^2 = \frac{0.45\nu}{U_e^6} \int_0^x U_e^5 dx'$$

(E06) *Thwaites pressure-gradient and momentum-thickness Reynolds number*

$$\lambda = \frac{\theta^2}{\nu} \frac{dU_e}{dx}, Re_\theta = \frac{U_e \theta}{\nu}$$

(E07) *Von Karman momentum-integral equation*

$$\frac{d\theta}{dx} + (2 + H) \frac{\theta}{U_e} \frac{dU_e}{dx} = \frac{C_f}{2}$$

4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the local Tu ; natural/TS onset integrates the envelope amplification rate of Drela & Giles in Re_θ and triggers at N_{crit} ; envelope; separation-induced and cross-flow onsets use bubble and C1 criteria. The kernel takes the minimum effective onset across all four:

(E08) *Abu-Ghannam & Shaw bypass onset*

$$Re_{\theta t} = 163 + \exp \left[F(\lambda_\theta) - \frac{F(\lambda_\theta) Tu}{6.91} \right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \leq 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (low-Tu correlation surrogate)

$$N(x) = \int_{x_0}^x -\alpha_i(x') dx' \geq N_{crit}$$

(E11) Cross-flow criterion (swept wing, C1 on Re_theta2)

$$Re_{\theta 2} = k_{cf} Re_\theta \sin \Lambda \cos \Lambda \geq C_1$$

(E12) Separation-induced onset (laminar bubble)

$$\lambda \leq -0.09 \Rightarrow Re_{\theta t, sep} = \max(Re_{\theta t}^{floor}, 0.7 Re_{\theta t, bp})$$

(E13) Unified minimum-onset transition kernel

$$Re_{\theta t}^* = \min(a_{TS} Re_{\theta t}^{TS}, a_{BP} Re_{\theta t}^{BP}, a_{SEP} Re_{\theta t}^{SEP}, a_{CF} Re_{\theta t}^{CF})$$

(E14) Transition trigger

$$Re_\theta(x_t) \geq Re_{\theta t}^*$$

4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x) = 1 - \exp[-0.412\xi^2], \xi = \frac{x - x_t}{\lambda_{tr}}$$

(E16) Transition-length scale

$$\lambda_{tr} \sim \frac{\nu}{U_e} C_{len} Re_{\theta t}^{0.8}$$

(E17) Intermittency-weighted property blend

$$\phi = (1 - \gamma)\phi_{lam} + \gamma\phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx} = C_E - \frac{\theta H_1}{U_e} \frac{dU_e}{dx}, C_E = 0.0306(H_1 - 3)^{-0.6169}$$

(E19) Ludwig-Tillmann skin-friction law

$$C_f = 0.246 \times 10^{-0.678H} Re_\theta^{-0.268}$$

4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2 \frac{\theta_{TE}}{c} \left(\frac{U_{e,TE}}{U_\infty} \right)^{(H_{TE}+5)/2}$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r \frac{\gamma - 1}{2} M_e^2 \left[1 - \left(\frac{u}{U_e} \right)^2 \right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e \left(1 + r \frac{\gamma - 1}{2} M_e^2 \right), r \approx Pr^{1/3}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_\infty U_\infty \bar{c}}{\mu_\infty} = \frac{U_\infty \bar{c}}{\nu_\infty}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

Capability	XFOIL / e ^N codes	RANS γ -Re _{θ} (CFD)	LES / DNS	UTSS (this work)
Natural / TS transition	Yes	Indirect	Yes	Yes (envelope e ^N)
Bypass (free-stream Tu)	No	Yes	Yes	Yes (AGS)
Separation-induced	Limited	Yes	Yes	Yes (bubble criterion)
Cross-flow (3-D swept)	No	Add-on only	Yes	Yes (C1, built-in)
Single universal calibration	n/a	Needs re-tuning	n/a	Yes (one constant set)
3-D wing capability	2-D only	Yes	Yes	Yes (strip + cross-flow)
Robustness / convergence	Can fail in sep.	Stiff, costly	Very costly	Robust, direct
Typical CPU cost	seconds	hours	days–weeks	< 1 second
Suited to design loops	Partly	No	No	Yes

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally by taking the minimum effective onset Reynolds number across four co-resident mechanisms, each modulated by a calibration weight, and feeds a single intermittency closure. Unlike e^N codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. The separation and cross-flow branches are carried and calibrated but are not selected at any condition reported here — the novelty claim.

6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

6.1 Aircraft and wing geometry (input)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m ²
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Table 3. Geometry definition (input).

6.2 Flight conditions (input)

case	altitude_m	mac_h	U_inf_ms	rho_kgm3	mu_Pas	T_inf_K	Tu_pct	alpha_deg	Re_MAC	nu_m2s	q_inf_Pa
CRUISE (FL360, M0.42)	11000.0	0.42	123.93	0.3639	1.422e-05	216.65	0.07	1.5	6414000.0	3.907e-05	2794.6
CLIMB (FL100, M0.30)	3000.0	0.3	98.57	0.9093	1.694e-05	268.66	0.9	3.5	10700000.0	1.863e-05	4417.8

Table 4. Flight / flow conditions (input).

Flight-condition comparison: cruise vs climb (flow_conditions.csv)

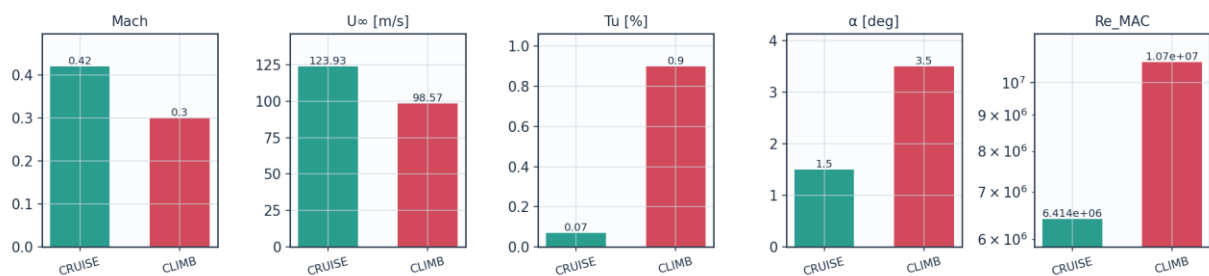


Fig. 8a. Cruise vs climb flight-condition comparison (flow_conditions.csv).

6.3 Fluid (material) properties (input)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K

Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Table 5. Air properties (input).

6.4 Solver settings (input)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility	none - incompressible panel solution
Laminar BL closure	Thwaites integral method
Transition model	UTSS unified 4-mechanism kernel
mechanisms	TS/natural(AGS@Tu=0.03%) bypass(AGS@Tu) separation crossflow(C1)
Intermittency closure	Narasimha universal (gamma)
Turbulent BL closure	Head entrainment + Ludwig-Tillmann Cf
Separation criterion	Thwaites lambda<=-0.09 / H>2.6 turbulent
Drag integration	Squire-Young far-wake
Marching scheme	explicit, arc-length stepping
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Table 6. Solver configuration (input).

6.5 Universal calibration constants (input)

constant	value	role	calibration_source
A_TS	1.0	TS/natural onset weight	validated on Schubauer-Skramstad (NACA Rep.909)
A_BP	1.0	bypass onset weight	validated on ERCOFTAC T3A/T3B
A_SEP	1.0	separation-induced weight	Mayle (1991) bubble correlation
A_CF	1.0	crossflow weight	Arnal C1 criterion
N_crit	from Tu	e^N amplification factor	Mack correlation N=-8.43-2.4 ln(Tu); 9.00 at Tu=0.07%
N_floor	0.5	lower clamp on N_crit at high Tu	clamp; bypass governs there
C_len	6.5	Narasimha transition-length scale	Dhawan-Narasimha (1958)
CF_C1	150.0	crossflow critical Re_theta2	Arnal et al. C1 criterion
CF_ratio	0.47	theta2/theta surrogate for crossflow	calibrated, not validated
sep_floor	120.0	min separation onset Re_theta	calibration floor

Table 7. Calibration constants and their sources (input).

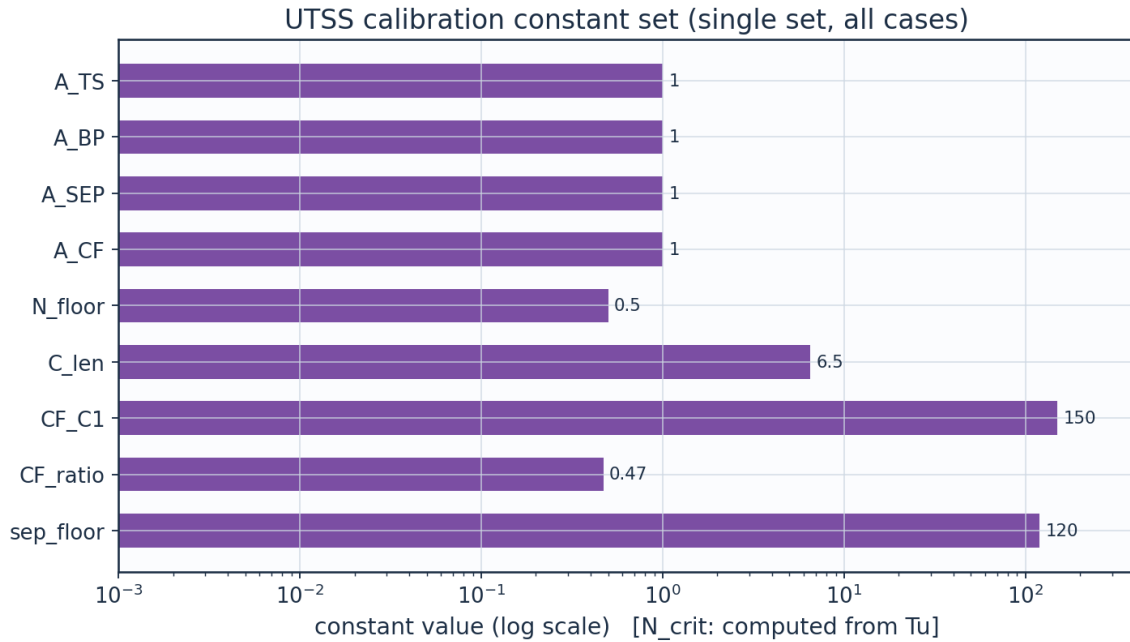


Fig. 8b. UTSS universal calibration constant set (calibration_constants.csv).

7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

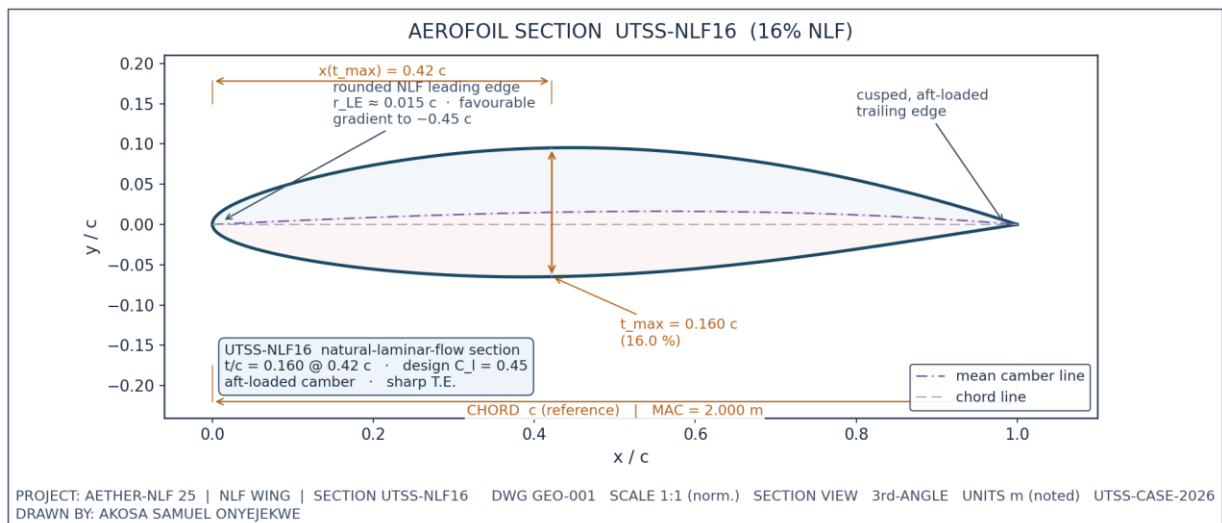


Fig. 1. UTSS-NLF16 aerofoil section (dimensioned).

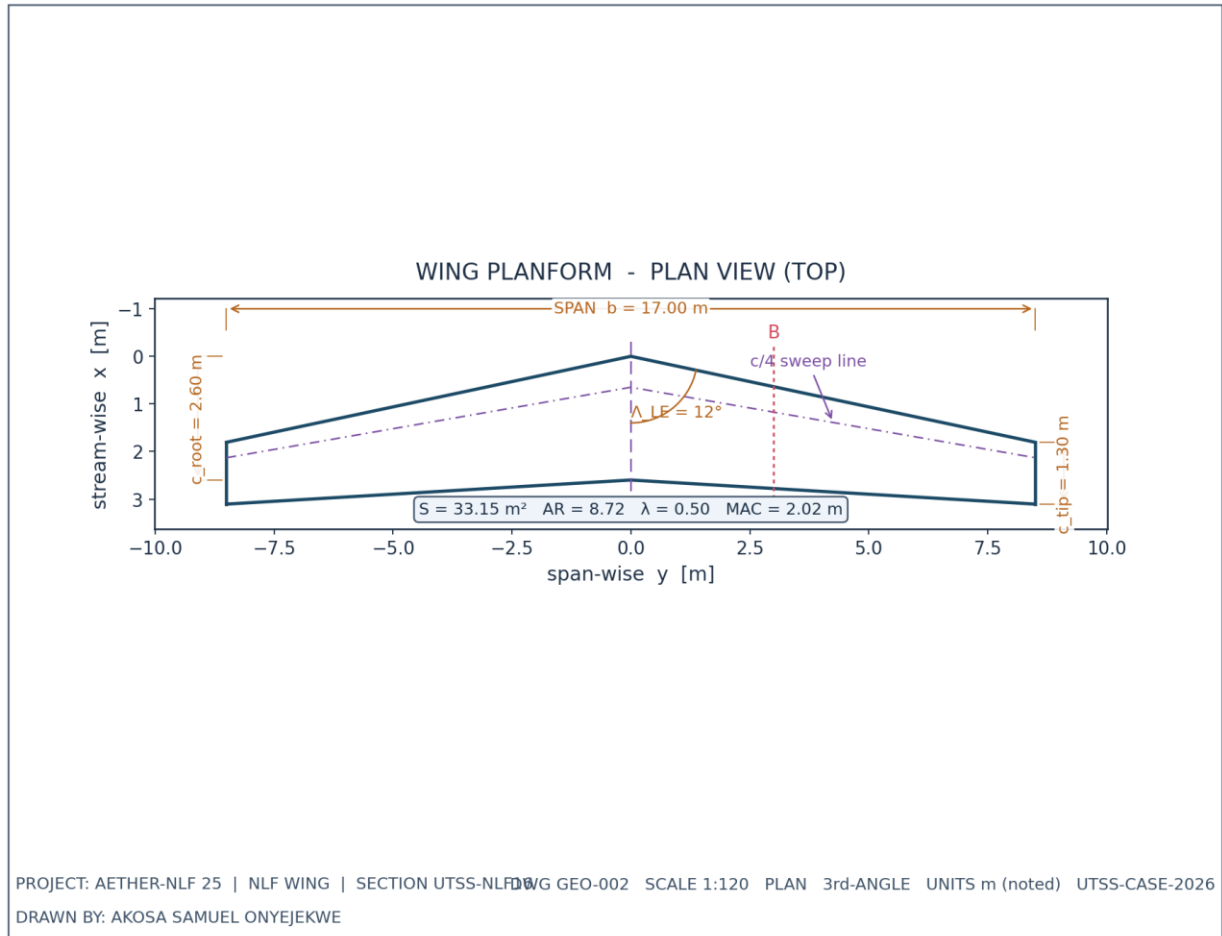


Fig. 2. Wing planform — plan (top) view, dimensioned.

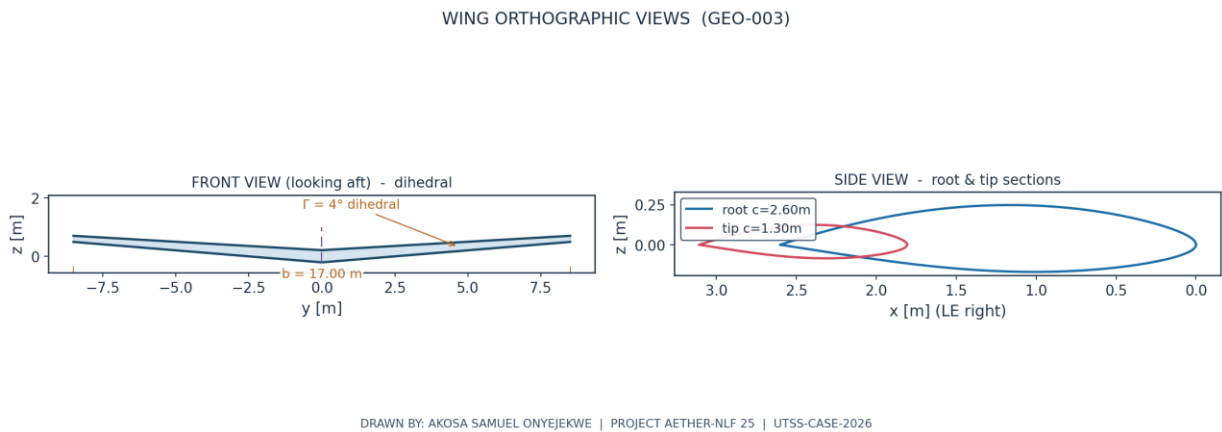
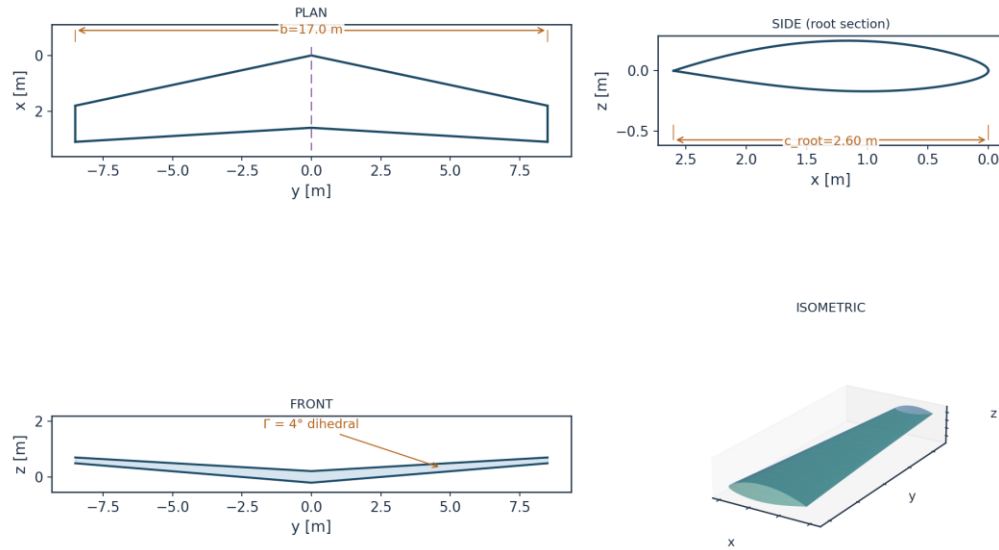


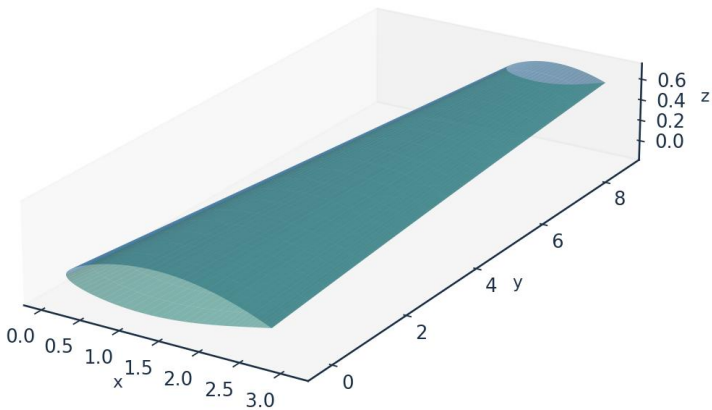
Fig. 3. Front and side orthographic views.



All dimensions in metres unless noted | Scale 1:120 | UTSS-CASE-2026 | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 4. Full third-angle orthographic projection sheet.

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 5. Isometric view of the wing.

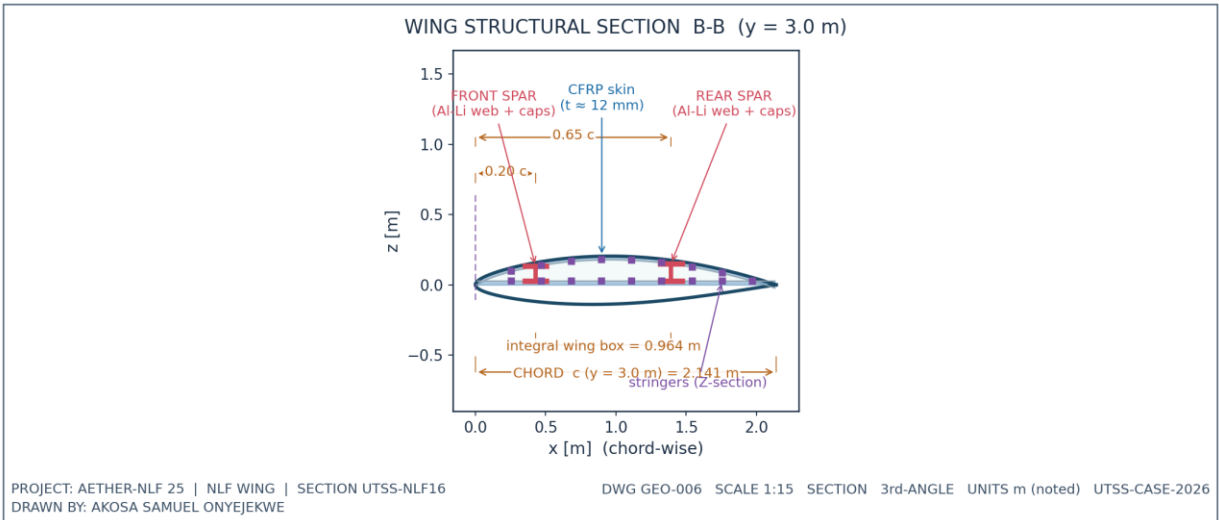


Fig. 6. Structural sectional view B-B.

7.1 Geometry data (from CSV)

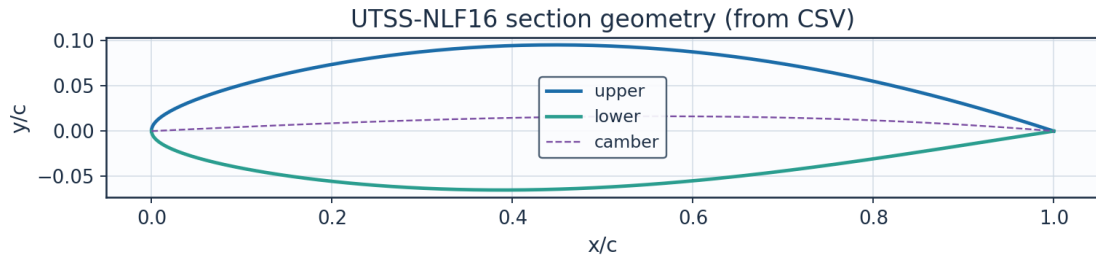


Fig. 7. Section geometry plotted from airfoil_UTSS-NLF16.csv.

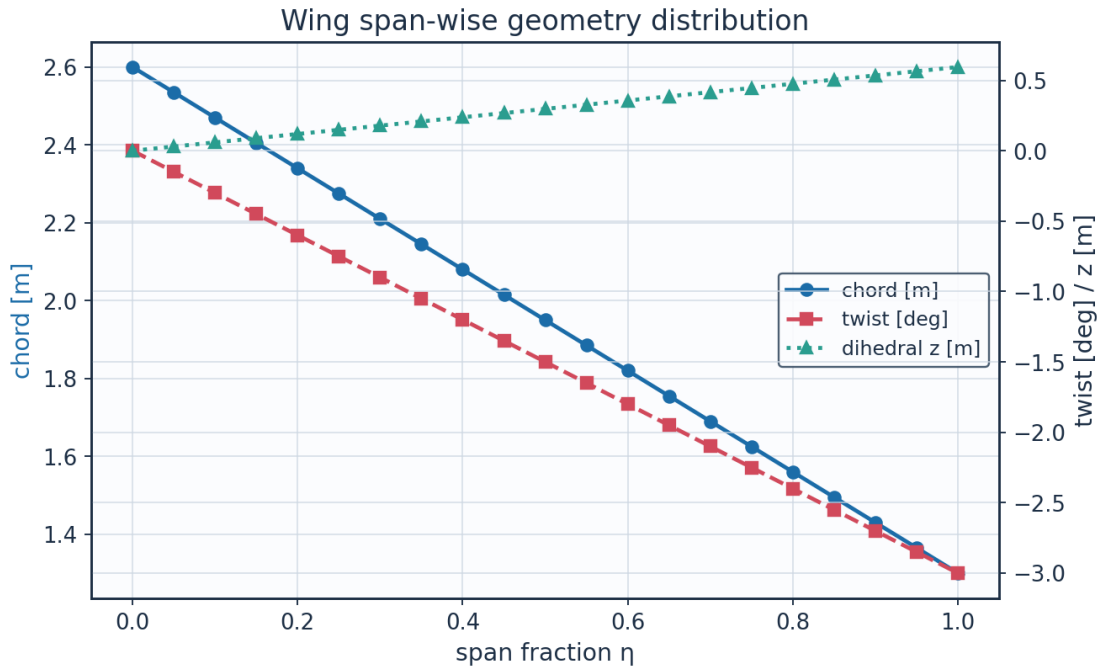


Fig. 8. Span-wise geometry from wing_planform.csv.

eta	y_m	chord_m	x_le_m	x_te_m	twist_deg	z_dihedral_m	Re_local
0.0	0.0	2.6	0.0	2.6	-0.0	0.0	8246203.012 995781
0.05	0.425	2.535	0.090336538 7097594	2.625336538 70976	-0.15	0.029718895 0759919	8040047.937 6708865
0.1	0.850000000 0000001	2.47	0.180673077 4195188	2.650673077 419519	-0.3	0.059437790 1519838	7833892.862 345993
0.15	1.275	2.405	0.271009616 1292782	2.676009616 1292783	-0.45	0.089156685 2279757	7627737.787 021098
0.2	1.700000000 0000002	2.34	0.361346154 8390376	2.701346154 8390376	- 0.600000000 0000001	0.118875580 3039677	7421582.711 696202
0.25	2.125	2.275	0.451682693 548797	2.726682693 548797	-0.75	0.148594475 3799596	7215427.636 371308
0.3	2.550000000 0000003	2.21	0.542019232 2585565	2.752019232 2585566	- 0.900000000 0000001	0.178313370 4559515	7009272.561 046414

0.35	2.975	2.145	0.632355770 9683159	2.777355770 968316	-1.05	0.208032265 5319434	6803117.485 721519
0.4	3.400000000 0000004	2.08	0.722692309 6780753	2.802692309 678076	- 1.200000000 0000002	0.237751160 6079354	6596962.410 396625
0.45	3.825	2.015	0.813028848 3878346	2.828028848 387835	-1.35	0.267470055 6839273	6390807.335 07173
0.5	4.25	1.95	0.903365387 097594	2.853365387 097594	-1.5	0.297188950 7599192	6184652.259 7468365
0.55	4.675000000 000001	1.885	0.993701925 8073536	2.878701925 807354	-1.65	0.326907845 8359112	5978497.184 421942
0.600000000 0000001	5.100000000 0000005	1.82	1.084038464 517113	2.904038464 5171127	- 1.800000000 0000005	0.356626740 9119031	5772342.109 097046
0.65	5.525	1.755	1.174375003 2268725	2.929375003 2268724	-1.95	0.386345635 987895	5566187.033 772152
0.700000000 0000001	5.95	1.69	1.264711541 9366315	2.954711541 936632	-2.1	0.416064531 0638869	5360031.958 447257
0.75	6.375	1.625	1.355048080 6463912	2.980048080 646391	-2.25	0.445783426 1398788	5153876.883 122363
0.8	6.800000000 000001	1.56	1.445384619 3561506	3.005384619 35615	- 2.400000000 0000004	0.475502321 2158709	4947721.807 797468
0.850000000 0000001	7.225	1.495	1.535721158 06591	3.030721158 06591	- 2.550000000 0000003	0.505221216 2918628	4741566.732 472573
0.9	7.65	1.43	1.626057696 7756692	3.056057696 775669	-2.7	0.534940111 3678547	4535411.657 14768
0.95	8.075000000 000001	1.365	1.716394235 485429	3.081394235 485429	-2.85	0.564659006 4438467	4329256.581 822785
1.0	8.5	1.3	1.806730774 195188	3.106730774 195188	-3.0	0.594377901 5198385	4123101.506 49789

Table 8. Wing planform stations (wing_planform.csv).

Lofted wing sections (from wing_sections_3d.csv)

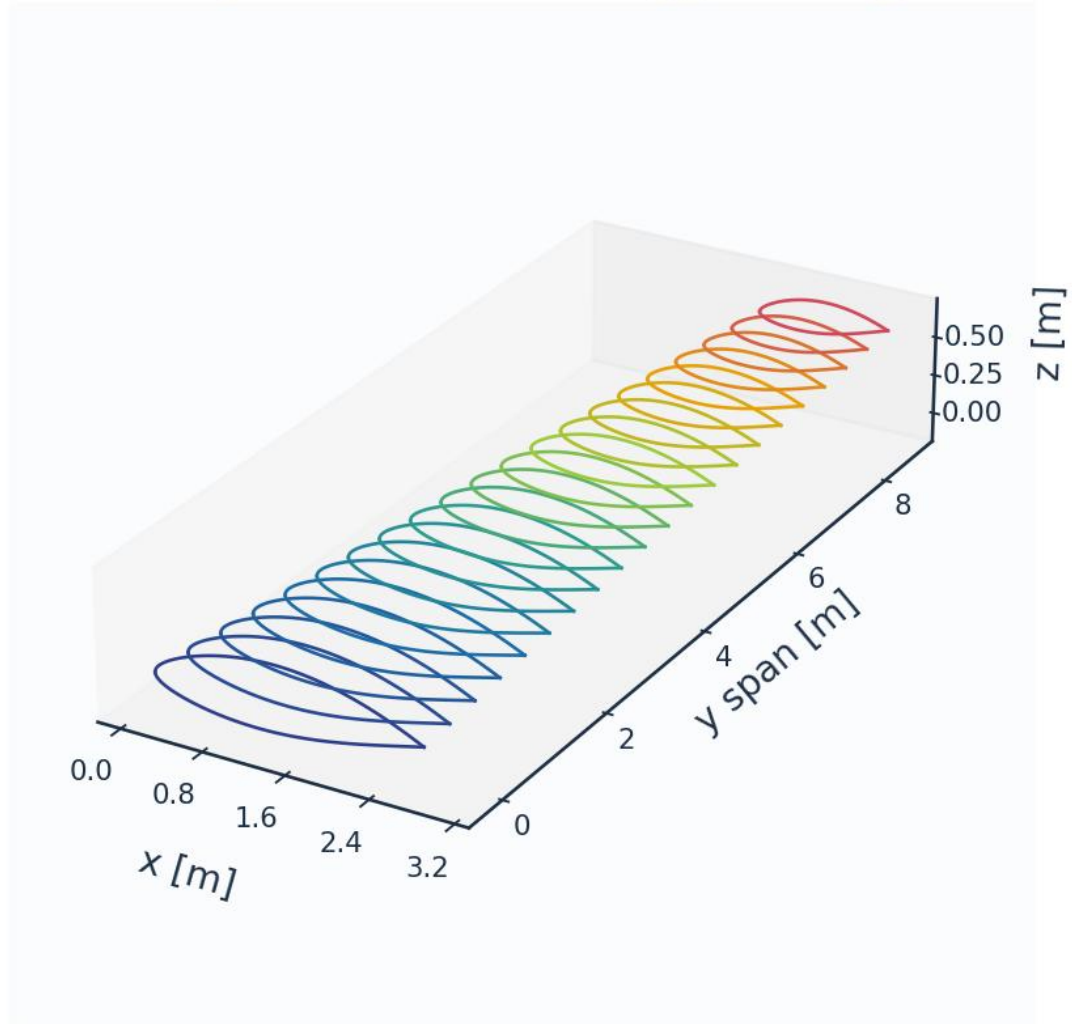


Fig. 8c. Lofted 3-D wing sections (wing_sections_3d.csv).

8. Mesh / Discretisation

The surface is discretised with cosine-clustered streamwise nodes; a wall-normal reconstruction grid (first-cell $y^+ \approx 1$) supports boundary-layer profile recovery. A mesh-independence study confirms convergence.

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y_1	6.51	micron
Target wall y^+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	mm (xc)
Max streamwise spacing	9.927	mm (xc)

LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_{τ} (TE)	4.800	m/s
Max cell aspect ratio	1524	-
Grid orthogonality (min)	88.5	deg
Mesh type	structured C-type (surface) + normal reconstruction	-

Table 9. Mesh metrics.

n_surface_panels	Cl	Cd	x_tr_upper_c	dCd_pct
120	0.4996087807380917	0.0044861077973023	0.4861367093300009	
180	0.5010076226503911	0.0048183870669946	0.4553399055769377	7.40684987311746
260	0.5018338859371162	0.0049596252583119	0.4445436762963268	2.931233820642909
360	0.5023435792157286	0.0050600731041123	0.442184134913257	2.0253111993098427
480	0.5026750206350319	0.0051339915261796	0.4367159348591741	1.460817275688342

Table 10. Mesh-independence study.

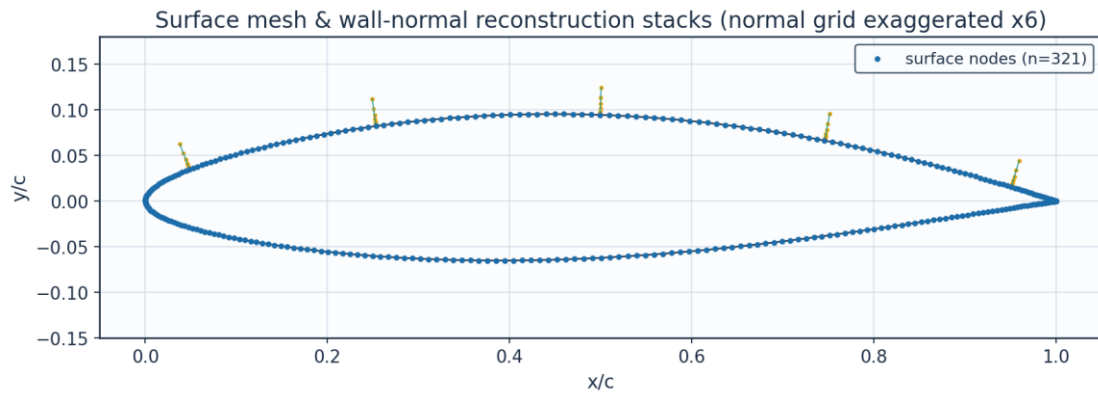


Fig. 9. Surface mesh and wall-normal stacks.

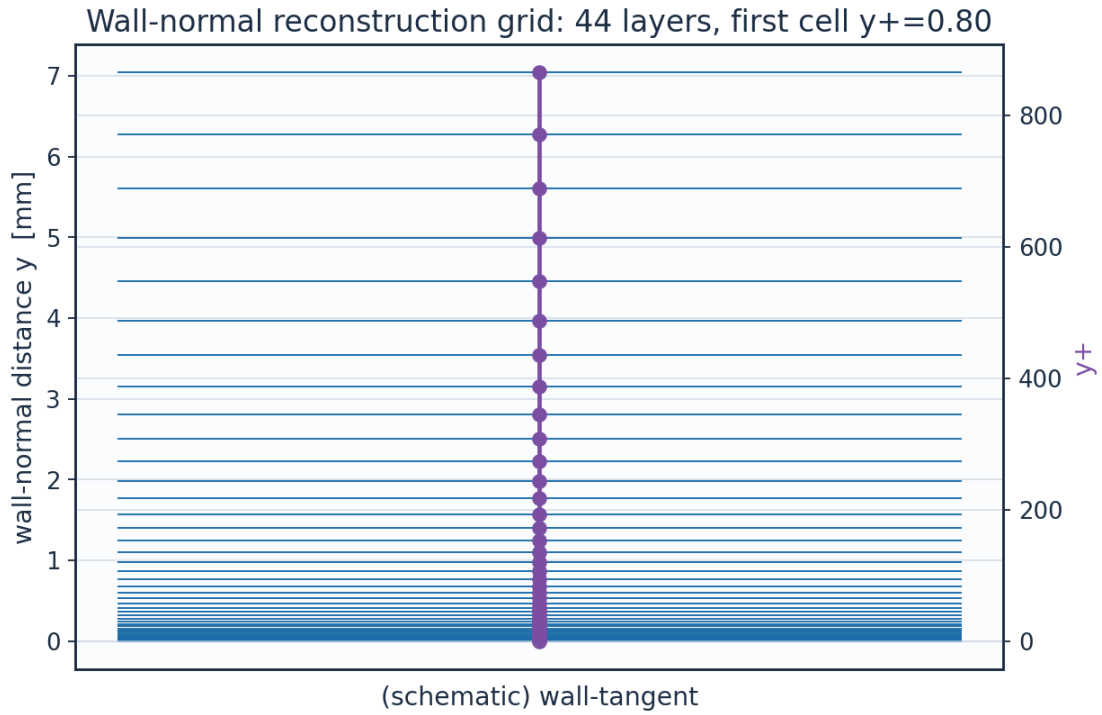


Fig. 10. Wall-normal reconstruction grid / y^+ .

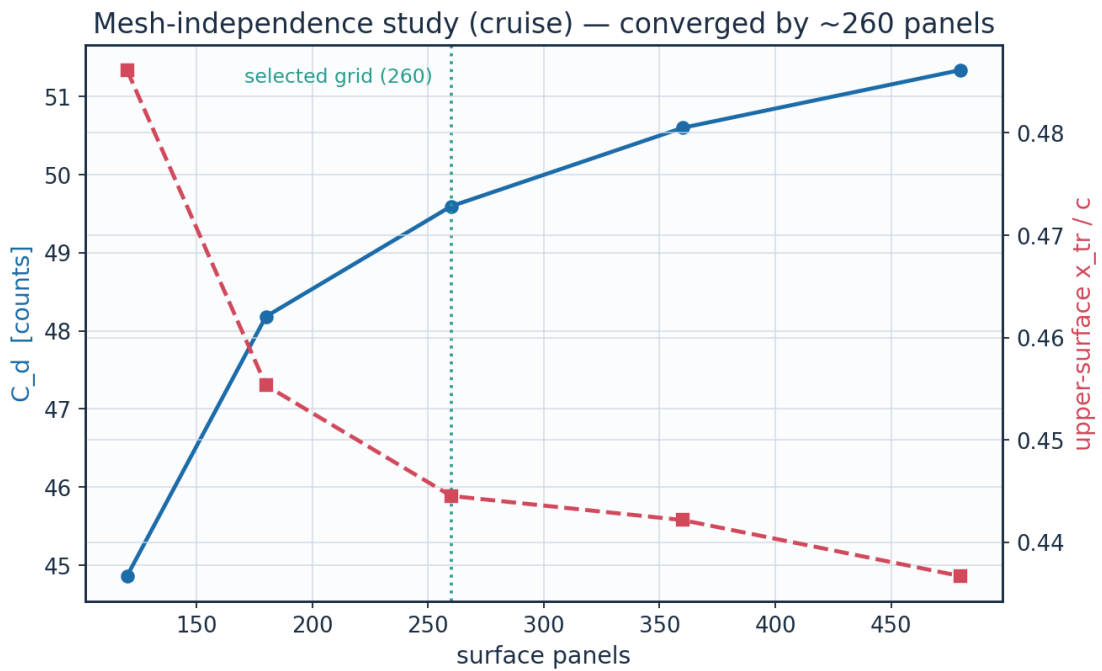


Fig. 11. Mesh-independence convergence.

layer	y_m	y_{plus}	cell_dy_m	growth_ratio
0.0	0.0	0.0	6.5127391337287455e-06	1.0
2.0	1.380700696350494e-	1.696	7.731923899562769e-	1.12

	05		06	
4.0	3.112651649852554e-05	3.8234624	9.698925339611536e-06	1.1199999999999997
6.0	5.2852109259255385e-05	6.4921512345600005	1.2166331946008715e-05	1.12
8.0	8.010469281831492e-05	9.839754508632067	1.526144679307334e-05	1.1199999999999997
10.0	0.0001142903336347	14.038988055628067	1.9143958857231197e-05	1.1199999999999997
12.0	0.0001571728014749	19.30650661697985	2.401418199051081e-05	1.12
14.0	0.0002109645691337	25.91408190033953	3.012338988889677e-05	1.12
16.0	0.0002784409624848	34.20262433578592	3.778678027663213e-05	1.1200000000000001
18.0	0.0003630833503045	44.599771966809854	4.739973717900734e-05	1.1199999999999997
20.0	0.0004692587615855	57.64195395516631	5.945823031734682e-05	1.12
22.0	0.0006024451974963	74.00206704136062	7.458440411007986e-05	1.12
24.0	0.0007695142627029	94.52419289668278	9.355867651568416e-05	1.12
26.0	0.000979085698098	120.26714756959888	0.0001173600038212	1.1200000000000006
28.0	0.0012419721066577	152.55910991130486	0.0001472163887934	1.1200000000000008
30.0	0.0015717368175549	193.0661474727408	0.0001846682381024	1.1199999999999997
32.0	0.0019853936709044	243.87817538980616	0.0002316478378757	1.1200000000000003
34.0	0.002504284827746	307.6167832089729	0.0002905790478312	1.1199999999999999
36.0	0.0031551818948881	387.57049285733575	0.0003645023575995	1.12
38.0	0.0039716671759111	487.864426240242	0.0004572317573729	1.12
40.0	0.0049958663124264	613.6731362757596	0.0005735515164485	1.1200000000000014
42.0	0.0062806217092712	771.4875821443129	0.000719463022233	1.1199999999999997

Table 11. Wall-normal grid (bl_normal_grid.csv, sampled).

(table sampled to 22 rows; full data in 02_mesh/bl_normal_grid.csv)

9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

9.1 Transition prediction summary

case	surface	s_tr_c	x_tr_c	Re_x_tr	Re_theta_at_onset	mechanism	laminar_run_pct	Cf_te
CRUISE	upper	0.469	0.445	3008000.0	1234.0	TS-natural	44.5	0.0002
CRUISE	lower	0.587	0.578	3766000.0	1277.1	TS-natural	57.8	0.00025
CLIMB	upper	0.047	0.025	506900.0	511.4	bypass	2.5	0.00014
CLIMB	lower	0.159	0.157	1706000.0	603.7	bypass	15.7	0.00019

Table 12. Transition prediction summary (transition_summary.csv).

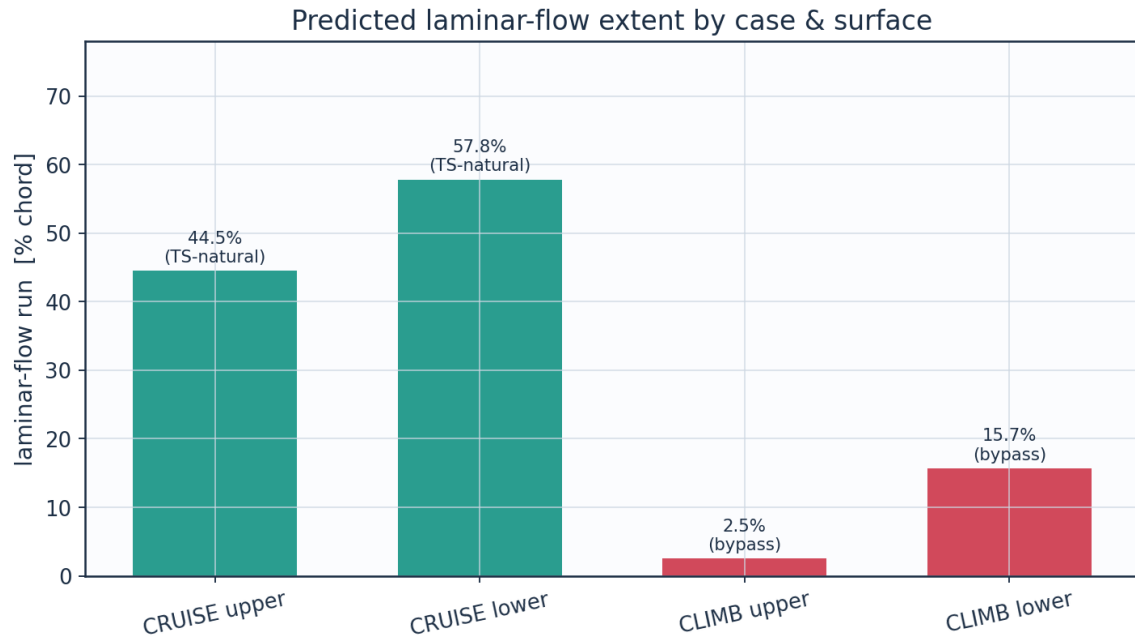


Fig. 12a. Predicted laminar-flow extent by case and surface (transition_summary.csv).

9.2 Integrated forces and drag breakdown

quantity	value	unit
Section lift coefficient Cl	0.5018	-
Section profile drag Cd	0.00496	-
Section Cd (counts)	49.6	counts
Section L/D	101.2	-
Wing lift (3D est, e=0.85)	41842.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Table 13. Integrated forces.

configuration	Cd_counts	mean_laminar_pct
NLF (UTSS predicted transition)	49.6	52.8
Fully turbulent (LE trip)	87.7	0.0
Viscous drag reduction	43.4	

Table 14. NLF vs fully-turbulent drag.

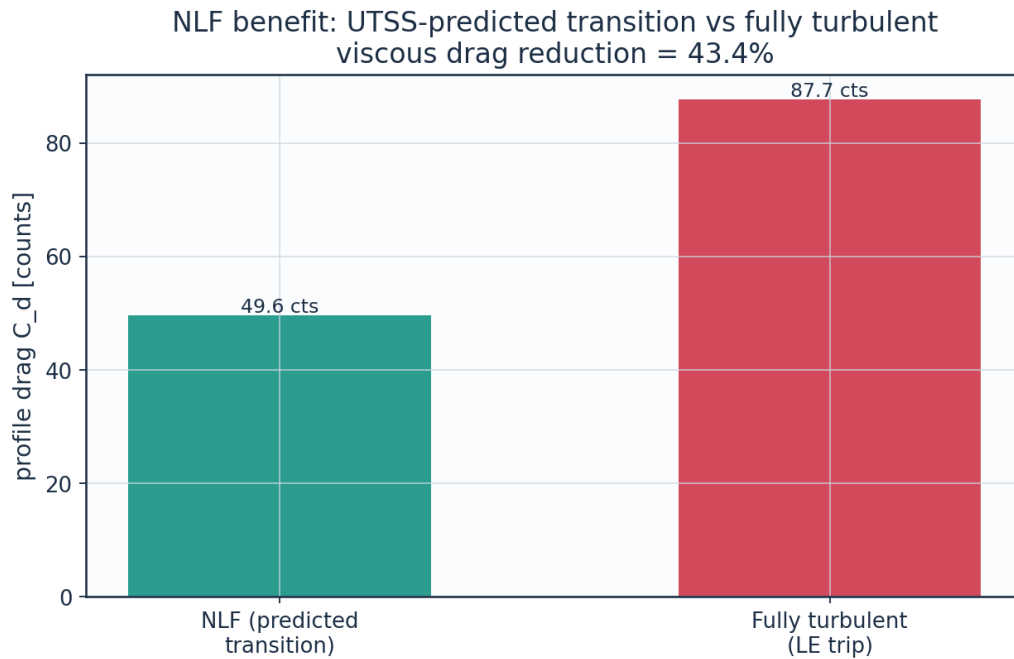


Fig. 12. Drag benefit of predicted laminar flow vs fully-turbulent.

9.3 Surface distributions — cruise

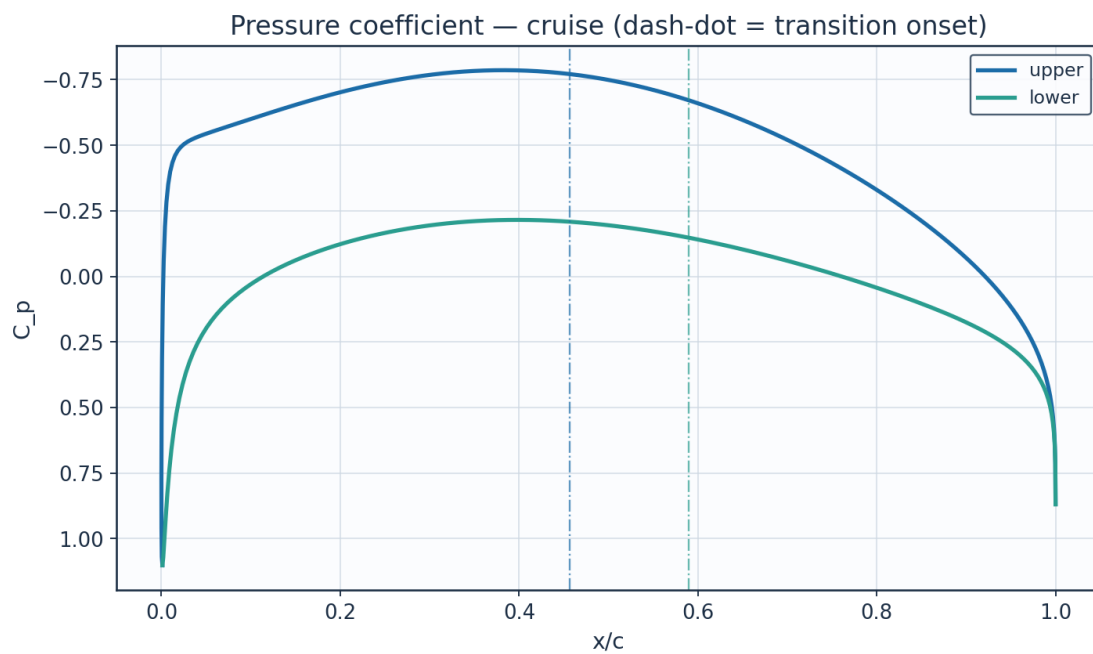


Fig. 13. Pressure coefficient C_p — cruise.

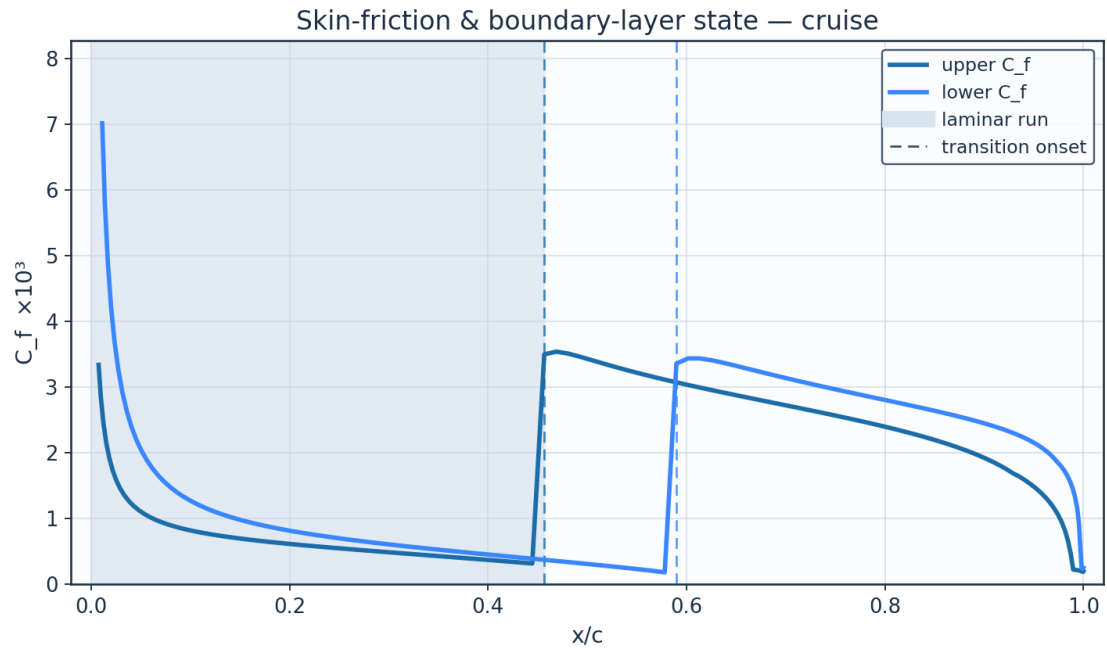


Fig. 14. Skin-friction C_f and laminar run — cruise.

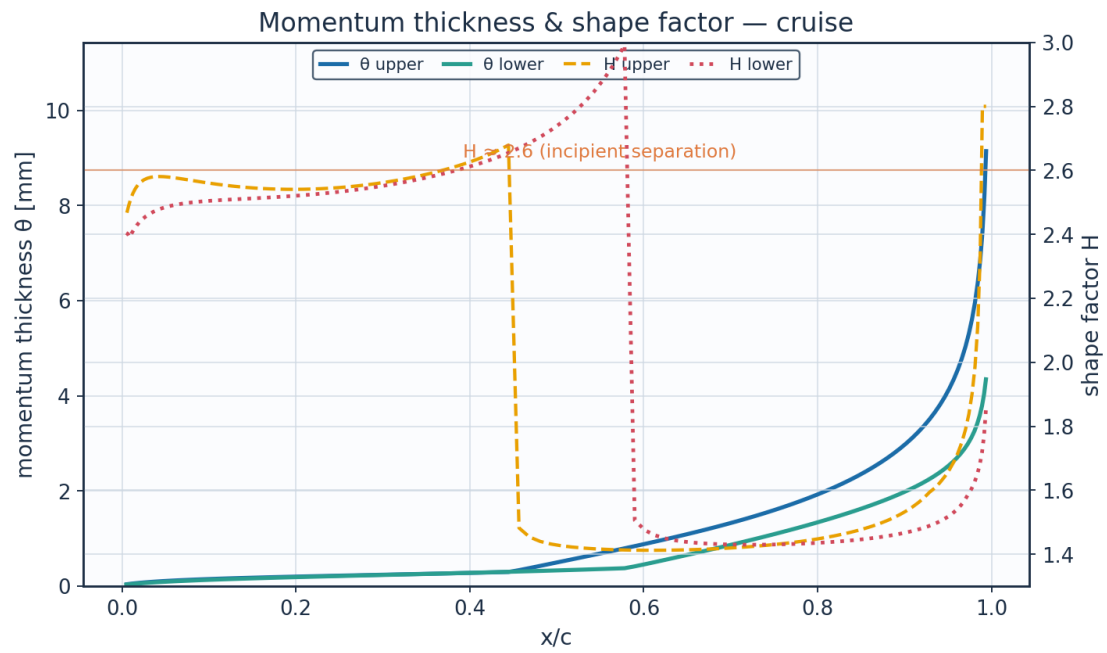


Fig. 15. Momentum thickness θ and shape factor H — cruise.

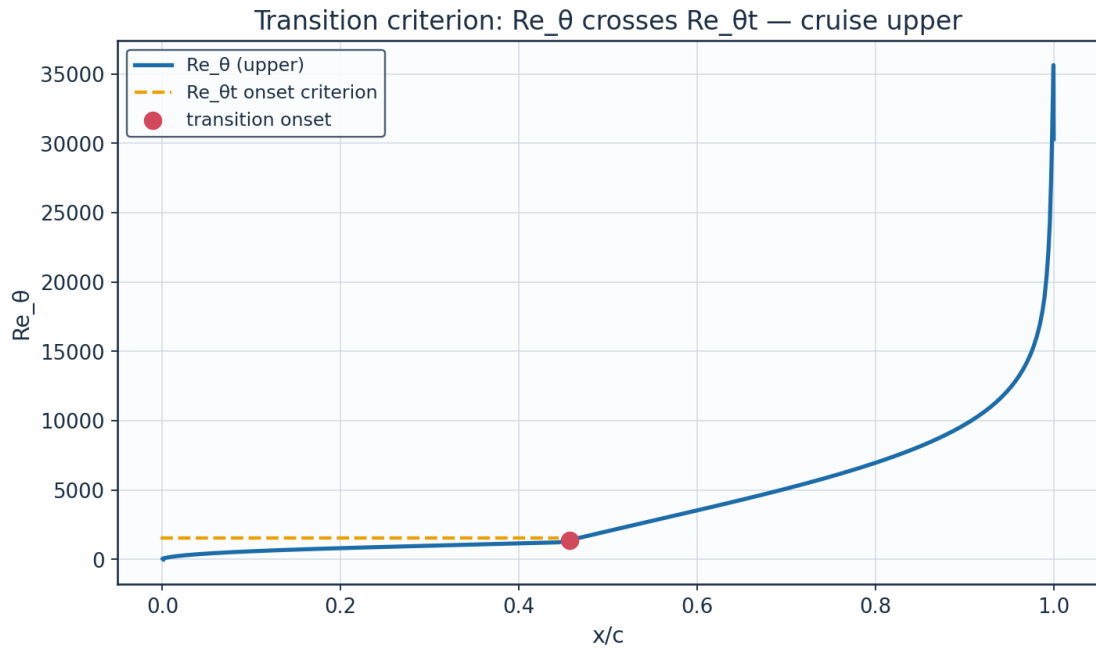


Fig. 16. Transition criterion Re_θ vs $Re_{\theta t}$ — cruise.

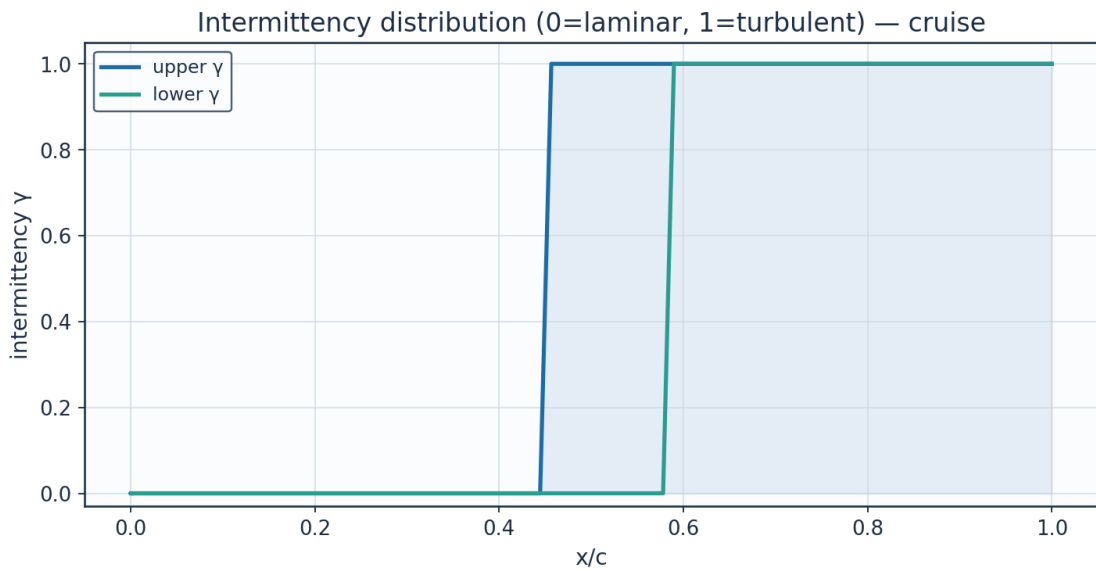


Fig. 17. Intermittency γ — cruise.

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.0011523200325028	0.0	0.0	1.1018664780939074	0.6743779759376758	0.0054416302825453	1e-05	2.6099999414506483	2549.4295711287004	0.0001725876462751	1569.314895047386	0.0	laminar
0.000242727	0.014097598	44712.17719	0.545321836	88.07774049	0.710709004	0.023652561	2.406649053	0.011498126	53.31515396	1569.314895	0.0	lami

1046013	6799438	636864	2263359	458249	4378694	2246563	519997	7313345	021425	047386		na r
0.0039984634929024	0.0303574508340593	96282.19328257376	-0.1889934015849139	134.13696378207896	1.0823659582155565	0.0351263884621607	2.4476972670491906	0.0047899370195188	120.58361487481666	1569.3148950473858	0.0	la mi na r
0.0123828259304446	0.052871121292092	167686.92296127492	-0.4342862195904694	146.32725910183973	1.1807307959357485	0.0553299604895908	2.526799032645004	0.0024617673195862	207.2010373148535	1569.3148950473862	0.0	la mi na r
0.0253172774881581	0.0831882375254351	263841.19035694306	-0.5041257659832884	149.6165255093602	1.2072722494368586	0.0780534449661155	2.570435671814969	0.0015833761627942	298.86700284754806	1569.3148950473862	0.0	la mi na r
0.0426830314241258	0.1217109964051874	386020.6097196828	-0.5346198063882681	151.03025162861334	1.218679761449772	0.1007487181096447	2.5805831844582143	0.001193288175888	389.4124231855982	1569.3148950473858	0.0	la mi na r
0.0643227586523211	0.1683869080802935	534058.7033695233	-0.5602079825700778	152.20640645138076	1.2281702844634186	0.1223924069362418	2.575118097854044	0.0009843198349802	476.75333039972	1569.314895047386	0.0	la mi na r
0.0900422957300214	0.2229351474645065	707064.8018171047	-0.5889179272140977	153.5153258571104	1.2387320995430495	0.1426396559680801	2.5641619639960385	0.000853860372772	560.4002868490857	1569.314895047386	0.0	la mi na r
0.1196122456699886	0.2849313998941607	903693.1416555132	-0.6215827965372597	154.99111243577204	1.250640384249194	0.1614919775048698	2.5530060519808826	0.0007616901542077	640.5662831735567	1569.3148950473858	0.0	la mi na r
0.1527694018013956	0.353848290318842	1122271.0914502463	-0.6567690814927153	156.56525350268606	1.263342302171124	0.1791518191450869	2.5445860710796717	0.0006896115453934	717.8321994860567	1569.3148950473858	0.0	la mi na r
0.1892179872734863	0.4290798210019653	1360876.6587546526	-0.6922076946556974	158.13484523358863	1.2760075109989126	0.1959464585740463	2.5406811264259623	0.0006284100151801	792.996597666261	1569.3148950473858	0.0	la mi na r
0.2286307839316745	0.5099588956971477	1617394.0707776232	-0.7253026000789115	159.5866938991054	1.2877226380431552	0.2122812285472066	2.542537498057847	0.0005729669422243	866.9910165721226	1569.3148950473858	0.0	la mi na r
0.2706503108546585	0.59577083610005	1889556.6398858647	-0.7533932601692048	160.80872570370516	1.2975833474841738	0.2286082830268302	2.551252922694963	0.000520175602888	940.822943359316	1569.3148950473858	0.0	la mi na r
0.3148902862023041	0.6857636986922375	2174979.491676561	-0.77395958212237	161.69756919660242	1.3047555236820514	0.2454058240566358	2.568066438141394	0.000467944770021	1015.5345047005782	1569.314895047386	0.0	la mi na r

			18									
0.3609	0.7791	24711	-	162.16	1.3085	0.2631	2.5946	0.0004	1092.1	1569.3	0.0	la
37646	55626	82.873	0.7848	46593	24525	65518	31894	14660	73225	14895		mi
56271	03463	46131	12690	46479	62894	02279	74833	96964	32259	04738		na
2	46	45	64728 3	48	28	33	47	78	23	62		r
0.4083	0.8751	27756	-	162.14	1.3083	0.2823	2.6342	0.0003	1171.7	1569.3	0.0	la
55390	39546	07.063	0.7842	11197	34582	86736	17595	58853	73817	14895		mi
17306	45295	36620	64986	75158	23115	55319	76994	08645	81596	04738		na
84	7	6	06502 61	2	92	37	85	74	86	6		r
0.4566	0.9728	30856	-	161.58	1.3038	0.3361	1.4820	0.0034	1389.8		1.0	tur
86447	85870	20.921	0.7712	10430	15261	11601	09187	97954	89383			bu
36747	56001	96654	56909	77768	57448	28550	74136	64545	74860			le
44	44	44	81593 15	6	22	37	04	64	64			nt
0.5054	1.0715	33985	-	160.46	1.2947	0.5180	1.4262	0.0034	2127.4		1.0	tur
58667	44176	27.236	0.7454	27537	91661	51459	56767	04617	21135			bu
99852	36863	81889	18708	07127	27753	16880	22215	20171	70919			le
56	5	24	83815 93	7	14	32	7	61	2			nt
0.5541	1.1702	37115	-	158.78	1.2812	0.7025	1.4152	0.0032	2854.7		1.0	tur
90867	45063	68.604	0.7070	78833	76947	07800	80641	00963	94411			bu
66523	74149	06482	56278	16317	32902	67190	22188	89648	67896			le
94	47	8	36423 4	87	69	91	4	58	9			nt
0.6023	1.2681	40219	-	156.57	1.2634	0.8917	1.4122	0.0030	3573.4		1.0	tur
99717	03309	37.436	0.6570	84251	48585	51246	05270	28544	02795			bu
62090	91279	53568	65002	17025	38026	08811	15801	54074	88480			le
82	1	5	49698 56	26	7	93	32	91	5			nt
0.6496	1.3642	43267	-	153.87	1.2416	1.0904	1.4135	0.0028	4294.2		1.0	tur
07126	23169	92.774	0.5967	22197	11915	88131	98187	76747	51633			bu
44050	95281	79366	88720	38401	64230	37427	37772	13254	29120			le
3	22	8	49912 81	55	58	67	6	03	4			nt
0.6953	1.4577	46232	-	150.71	1.2161	1.3032	1.4186	0.0027	5027.0		1.0	tur
47671	06209	85.129	0.5278	74710	55900	95756	40924	36180	46551			bu
73678	48466	50581	48430	46897	43119	42876	27371	23487	55698			le
39	56	1	52714 63	32	17	46	6	79	4			nt
0.7391	1.5476	49085	-	147.16	1.1875	1.5349	1.4272	0.0026	5781.1		1.0	tur
75615	61543	88.956	0.4519	68703	05712	67286	61765	00348	66881			bu
76998	81474	03955	65754	97203	10062	88717	25552	01137	78120			le
33	4	5	18730 16	18	5	1	62	03	7			nt
0.7806	1.6332	51799	-	143.27	1.1560	1.7909	1.4398	0.0024	6566.9		1.0	tur
71085	17905	40.926	0.3708	17466	75529	91530	37867	64180	00183			bu
14247	01870	63225	00761	98236	27906	31702	79685	07430	82474			le
59	1	7	96327 45	7	32	3	49	93	7			nt
0.8194	1.7135	54346	-	139.07	1.1222	2.0782	1.4572	0.0023	7396.8		1.0	tur
45102	36656	81.206	0.2858	64000	22814	09341	20283	22921	88301			bu
96803	13103	41017	15047	66342	49230	01257	85929	88915	59276			le
74	57	8	90891 36	4	3		54	93	8			nt
0.8551	1.7878	56702	-	134.61	1.0862	2.4058	1.4809	0.0021	8288.1		1.0	tur
43310	24762	94.594	0.1981	24371	02604	46987	62385	71178	86542			bu
57486	77398	42127	61236	0955	94725	70998	33162	14750	15180			le
8	5		85266 84		24	26	85	32	7			nt

0.8874 48376 11318 26	1.8553 46811 45599 4	58844 48.641 06943 6	- 0.1085 90218 31808 16	129.89 24515 20186 06	1.0481 16520 53658 03	2.7873 15605 08364 54	1.5138 79365 67366 1	0.0020 01713 99431 42	9265.6 60582 67043 3		1.0	tur bu le nt
0.9160 81229 91338 4	1.9154 35385 45516 16	60750 26.556 43811 1	- 0.0173 55583 78332 83	124.90 15687 55567 13	1.0078 44536 93447 35	3.2436 43751 77252 43	1.5614 19285 88537 44	0.0018 03363 41988 72	10368. 29559 67570 54		1.0	tur bu le nt
0.9408 01367 76385 24	1.9674 99407 22596 13	62401 53.669 20543 7	0.0759 25683 20532 58	119.58 35624 55573 84	0.9649 32957 43845 96	3.8106 23231 14421 4	1.6238 04616 45895 37	0.0015 85258 87029 78	11662. 02116 65796 34		1.0	tur bu le nt
0.9614 06511 28321 76	2.0110 30361 48001 73	63782 17.163 87019 8	0.1725 46860 87025 07	113.81 34551 80803 96	0.9183 73325 30298 68	4.5537 12660 46987 5	1.7160 67782 48861 84	0.0013 26071 01026 63	13263. 72651 40172 1		1.0	tur bu le nt

Table 15. Cruise upper-surface BL state (surface_cruise_upper.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_upper.csv)

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.0011 52320 03250 28	0.0	0.0	1.1018 66478 09390 74	0.6743 77975 93767 58	0.0054 41630 28254 53	1e-05	2.6099 99969 21184 96	2549.4 29436 44131 3	0.0001 72587 64627 51	1569.3 14895 04738 6	0.0	la mi na r
0.0067 20774 97236 91	0.0175 58980 96768 28	55690. 35452 34006 45	0.8272 66067 17411 17	61.869 95831 42899 5	0.4992 35518 88647 86	0.0360 07692 15972 1	2.4014 55110 01168 63	0.0108 31210 90974 53	57.013 99034 79811 6	1569.3 14895 04738 6	0.0	la mi na r
0.0168 97481 50526 17	0.0419 27102 59539 91	13297 6.6922 10909	0.5213 99709 95785 02	89.950 65248 46617 2	0.7258 21737 89804 29	0.0527 26068 01883 9	2.4275 91205 30258 88	0.0049 01321 98185 56	121.37 68971 16424 91	1569.3 14895 04738 58	0.0	la mi na r
0.0315 87190 34084 9	0.0742 42760 85277 24	23546 9.5685 52791 4	0.3249 25538 30279 69	104.06 54470 09190 68	0.8397 15572 00463 09	0.0725 75117 95132 31	2.4636 43616 82537 5	0.0029 17440 77671 5	193.28 61038 17425 9	1569.3 14895 04738 62	0.0	la mi na r
0.0506 50104 83468 89	0.1147 42805 66774 19	36392 0.1806 98816 6	0.1950 00250 86657 96	112.43 01677 32621 64	0.9072 11426 28481 72	0.0931 90090 93638 6	2.4852 76623 75723 77	0.0020 34017 67762 49	268.13 83431 65555 5	1569.3 14895 04738 58	0.0	la mi na r
0.0739 02667 92458 39	0.1632 86856 80577 34	51788 3.2963 74761 5	0.1010 10397 90928 31	118.11 26263 41395 54	0.9530 63811 66531 56	0.1137 27691 64392 47	2.4975 92677 31736 53	0.0015 55968 54585 14	343.77 07156 46582 97	1569.3 14895 04738 6	0.0	la mi na r
0.1011 19142 10955 74	0.2195 30525 04901 36	69626 6.6450 39893 18	0.0275 36118 51389 18	122.37 11244 77975 48	0.9874 26102 91000 77	0.1338 21916 27854 77	2.5047 42863 77289 4	0.0012 61789 87201 49	419.09 50417 27011 4	1569.3 14895 04738 62	0.0	la mi na r
0.1320 34052 45495 44	0.2829 94629 23130 39	89755 0.4477 80308 7	- 0.0328 99357 93687 48	125.76 58705 52814 48	1.0148 18683 48280 97	0.1533 64785 45435 72	2.5096 84230 40556 44	0.0010 62771 87861 11	493.62 23101 47081 3	1569.3 14895 04738 6	0.0	la mi na r
0.1663	0.3531	11199	-	128.55	1.0373	0.1724	2.5146	0.0009	567.24	1569.3	0.0	la

45511 63160 67	01330 53650 2	02.021 44727 8	0.0837 64019 94194 75	35542 38382 13	12810 66786 53	18483 25019 9	15367 27198 13	17441 05390 9	96495 35809 7	14895 04738 58		mi na r
0.2037 19371 58191 91	0.4291 98466 63705 09	13612 52.957 21369 96	- 0.1264 28412 92398 36	130.84 60216 58987 58	1.0558 10983 18064 64	0.1911 66084 86118 05	2.5212 76076 95081 2	0.0008 04138 11247 12	640.14 40243 22221	1569.3 14895 04738 58	0.0	la mi na r
0.2437 94056 30589 38	0.5105 78867 09950 84	16193 60.381 63382 23	- 0.1611 53110 05562 72	132.68 26339 36490 02	1.0706 30810 25559 77	0.2098 77106 12853 9	2.5311 47244 29188 25	0.0007 10567 15047 72	712.66 50796 81708 6	1569.3 14895 04738 6	0.0	la mi na r
0.2861 85839 50534 81	0.5964 96549 11452 82	18918 58.323 28838 8	- 0.1876 58005 00392 6	134.06 75651 33080 85	1.0818 05972 86100 68	0.2288 80253 37145 03	2.5456 04320 67113 5	0.0006 29264 75230 6	785.30 50224 47479 5	1569.3 14895 04738 62	0.0	la mi na r
0.3304 94258 79381 13	0.6861 79937 75148 77	21762 99.642 36291 36	- 0.2054 90229 40969 07	134.99 13376 06888 24	1.0892 59994 85907 5	0.2485 40393 16493 53	2.5660 50106 71399	0.0005 55430 31674 93	858.63 62292 99890 6	1569.3 14895 04738 6	0.0	la mi na r
0.3763 07324 00294 27	0.7788 41767 61081 37	24701 87.434 89202	- 0.2142 79201 03680 94	135.44 43196 35476 97	1.0929 15156 81893 88	0.2692 38193 42271 7	2.5940 40102 98206 88	0.0004 85799 24503 73	933.26 24477 55753	1569.3 14895 04738 6	0.0	la mi na r
0.4232 06194 57674 89	0.8736 85455 08345 47	27709 95.243 12299 02	- 0.2139 09854 12289 9	135.42 53140 60306 2	1.0927 61798 73626 6	0.2913 51474 23027 07	2.6321 61672 83077 6	0.0004 18018 10515 92	1009.7 72173 99793 6	1569.3 14895 04738 58	0.0	la mi na r
0.4707 69074 33519 84	0.9699 08166 57242 24	30761 76.786 73797 05	- 0.2046 21272 31479 29	134.94 64691 14025 58	1.0888 97946 04045 2	0.3152 37821 95105 41	2.6887 22664 98492 7	0.0003 50158 66484 31	1088.6 94857 01760 23	1569.3 14895 04738 62	0.0	la mi na r
0.5185 74193 40729 35	1.0667 01325 94151 12	33831 67.572 28672 46	- 0.1870 32340 06093 01	134.03 50378 60131 75	1.0815 43506 70730 07	0.3412 18707 81401 65	2.7783 12019 91752 9	0.0002 79823 93919 41	1170.4 62496 48701 88	1569.3 14895 04738 6	0.0	la mi na r
0.5662 01898 02471 88	1.1632 49726 11984 71	36893 82.075 53692 4	- 0.1621 00475 13805 48	132.73 23845 84627 28	1.0710 32253 72371 11	0.3695 66262 35430 63	2.9337 06254 59574 97	0.0002 02108 31513 69	1255.3 80972 38463 68	1569.3 14895 04738 58	0.0	la mi na r
0.6132 36022 35407 45	1.2587 30610 74833 58	39922 10.828 80882 6	- 0.1310 27712 20340 01	131.09 07600 19421 3	1.0577 85804 00955 16	0.5071 02572 89382 49	1.4588 27936 29857 4	0.0034 35597 47255 43	1701.2 73835 27867 3		1.0	tur bu le nt
0.6592 64843 41077 2	1.3523 14020 52265 4	42890 21.519 44244 1	- 0.0951 35109 81048 94	129.16 85290 40939 32	1.0422 75110 19128 1	0.6943 20722 35694 23	1.4356 92753 13303 2	0.0032 87265 72325 44	2295.2 13916 65701 28		1.0	tur bu le nt
0.7038 81995	1.4431 65382	45771 67.201	- 0.0557	127.02 45984	1.0249 75497	0.8839 42318	1.4300 24515	0.0031 22643	2873.5 45540		1.0	tur bu

09891 86	46492 15	97440 2	28294 73007 42	37240 29	64011 04	66921 08	13806 73	74369 69	38447			le nt
0.7466 87725 12314 54	1.5304 50847 76179 95	48540 03.227 71357 7	- 0.0139 73220 73816 02	124.71 27010 62973 32	1.0063 20542 68779 9	1.0770 16603 81495 58	1.4300 76740 93233 92	0.0029 75986 21008 99	3437.4 73916 84540 3		1.0	tur bu le nt
0.7872 90817 05456 05	1.6133 45372 85086 3	51169 12.874 84830 4	0.0292 06547 98627 41	122.27 59555 95344 84	0.9866 58174 69740 72	1.2750 37476 64739 95	1.4335 73855 48759 83	0.0028 43884 67649 93	3989.9 76547 47786 7		1.0	tur bu le nt
0.8253 11381 55209 68	1.6910 43098 74663 05	53633 40.267 68853 3	0.0732 09048 09547 94	119.74 17781 22673 57	0.9662 09617 11001 24	1.4795 37847 10843 95	1.4399 34381 39123 5	0.0027 20968 18110 75	4533.9 64438 22515 6		1.0	tur bu le nt
0.8603 84569 42783 87	1.7627 69288 93695 93	55908 28.239 09553 2	0.1178 16707 00889 07	117.11 67909 03076 12	0.9450 28305 65689 92	1.6926 92284 31211 48	1.4492 80068 08548 38	0.0026 01967 58198 8	5073.4 51415 98071 2		1.0	tur bu le nt
0.8921 65104 53399 4	1.8277 92963 07022 2	57970 58.399 69313 5	0.1632 58645 36008 6	114.38 07835 11247 22	0.9229 51160 18674 52	1.9183 56574 88258 4	1.4623 52717 14780 32	0.0024 80982 98247 72	5615.5 03851 90978 3		1.0	tur bu le nt
0.9203 32404 83917 28	1.8854 39395 75766 85	59798 90.779 27608 8	0.2103 00243 58521 76	111.47 77474 37841 64	0.8995 26241 85900 72	2.1638 48137 75419 77	1.4808 13334 82023 77	0.0023 50087 10955 03	6173.3 55543 00056 8		1.0	tur bu le nt
0.9445 95976 40102 88	1.9351 01792 46614 69	61374 00.858 26456 7	0.2604 12207 78335 69	108.29 96644 45970 52	0.8738 81939 60408 53	2.4433 90311 50479 4	1.5081 33184 91071 7	0.0021 96777 96208 65	6772.1 45586 50210 6		1.0	tur bu le nt
0.9647 00736 13354 88	1.9762 51652 32757 04	62679 12.434 56211 6	0.3161 38102 70176 09	104.65 22734 79346 2	0.8444 50739 52868 49	2.7859 51607 15581 26	1.5522 26970 95234 75	0.0019 98053 74438 88	7461.5 41575 36116 1		1.0	tur bu le nt
0.9804 31941 29791 1	2.0084 47529 88663 1	63700 25.412 45964 3	0.3819 99182 65001 51	100.17 04201 74896 92	0.8082 86170 79479 88	3.2572 16603 59362 8	1.6237 73486 08361 84	0.0017 33816 38768 75	8350.1 15117 04730 8		1.0	tur bu le nt

Table 16. Cruise lower-surface BL state (surface_cruise_lower.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_lower.csv)

9.4 Surface distributions — climb (off-design, elevated T_u)

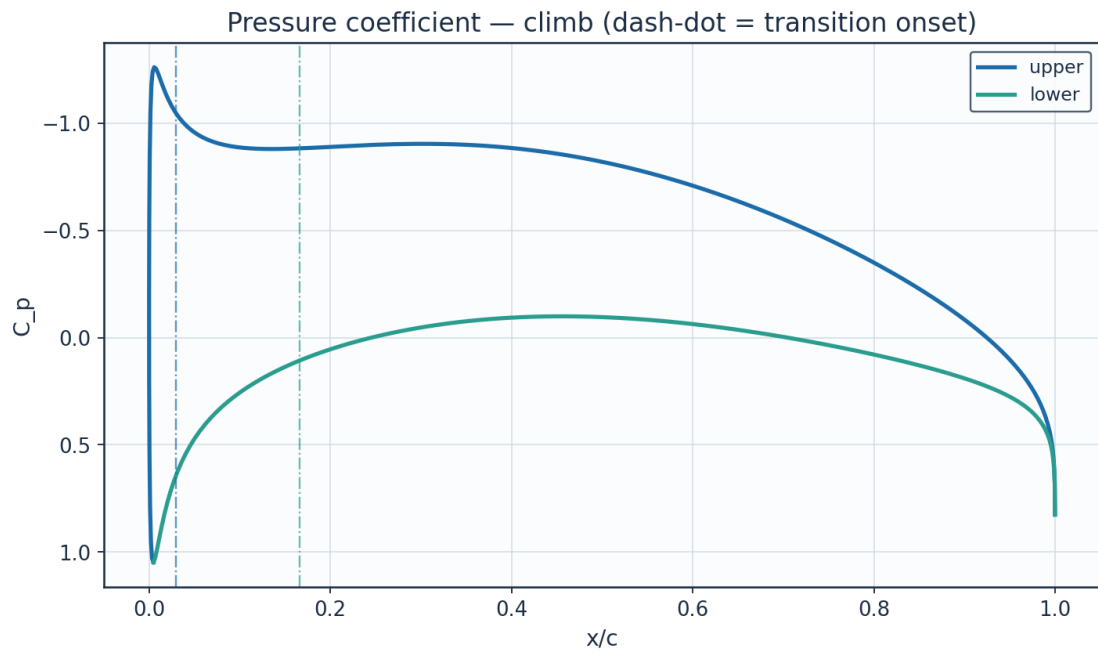


Fig. 18. Pressure coefficient C_p — climb.

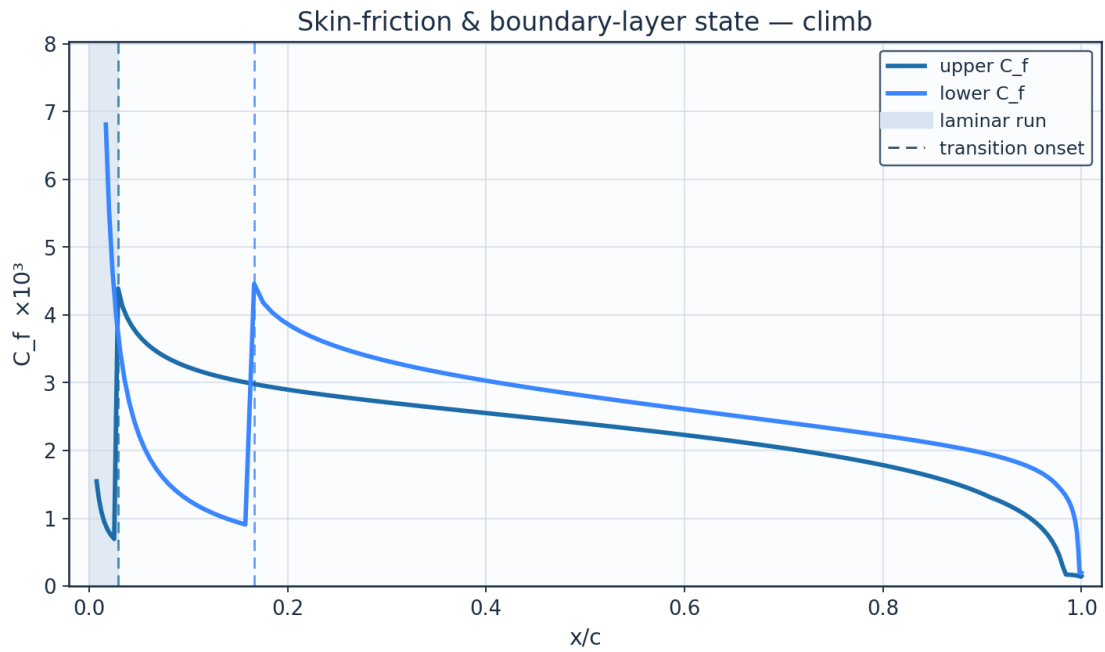


Fig. 19. Skin-friction C_f and laminar run — climb.

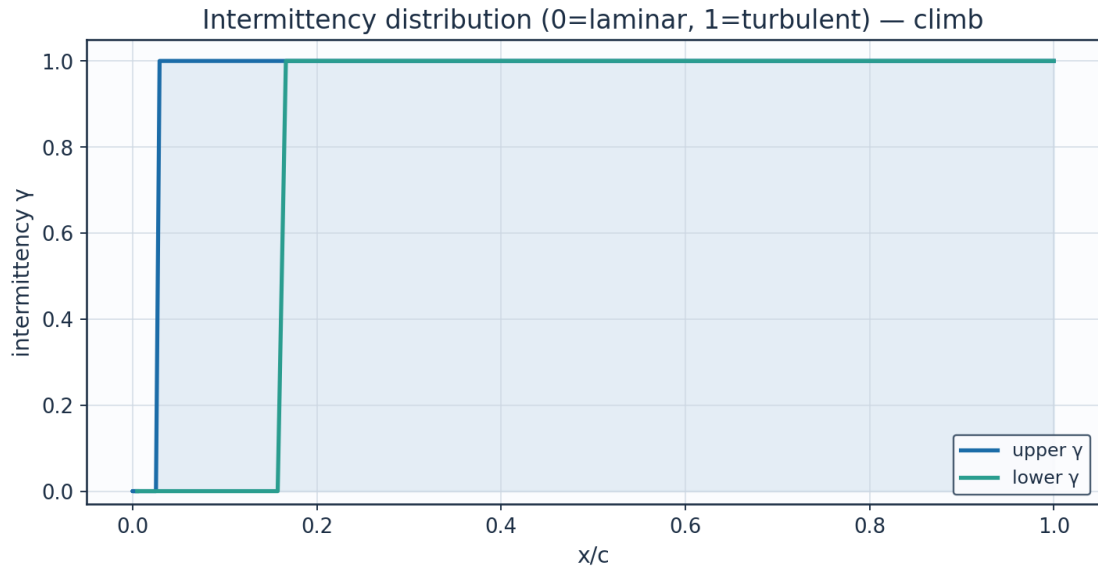


Fig. 20. Intermittency γ — climb.

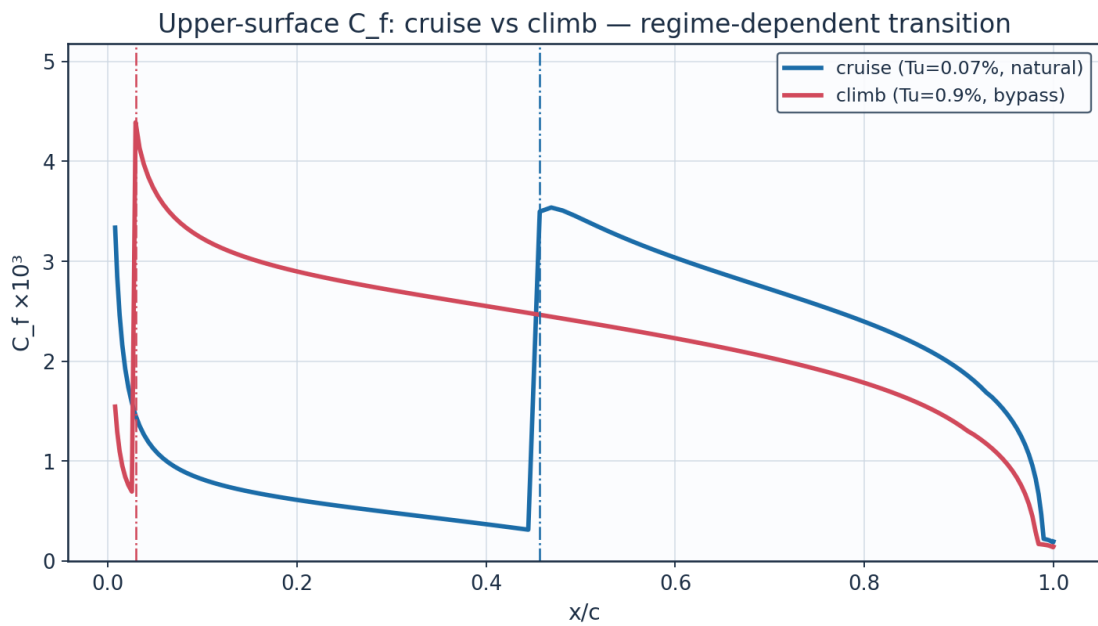


Fig. 21. C_f cruise vs climb — regime-dependent transition.

9.5 Aerodynamic polars

alpha_deg	Cl	Cd	L_over_D	xtr_upper_c	xtr_lower_c
-3.0	-0.1086	0.00545	-19.9	0.661	0.194
-2.0	0.0271	0.00485	5.6	0.638	0.33
-1.0	0.1628	0.00453	35.9	0.614	0.423
0.0	0.2985	0.00446	67.0	0.566	0.507
1.0	0.4341	0.00472	91.9	0.493	0.554
2.0	0.5695	0.00521	109.3	0.396	0.602
3.0	0.7048	0.00611	115.3	0.249	0.636

4.0	0.8399	0.00724	116.0	0.09	0.659
5.0	0.9746	0.00812	120.1	0.033	0.693
6.0	1.1091	0.00886	125.2	0.018	0.715
7.0	1.2432	0.00964	129.0	0.012	0.747
8.0	1.3769	0.01052	130.9	0.01	0.767

Table 17. Aerodynamic polar (aero_polar.csv).

Aerodynamic polars (UTSS, cruise Re) — AETHER-NLF 25 section

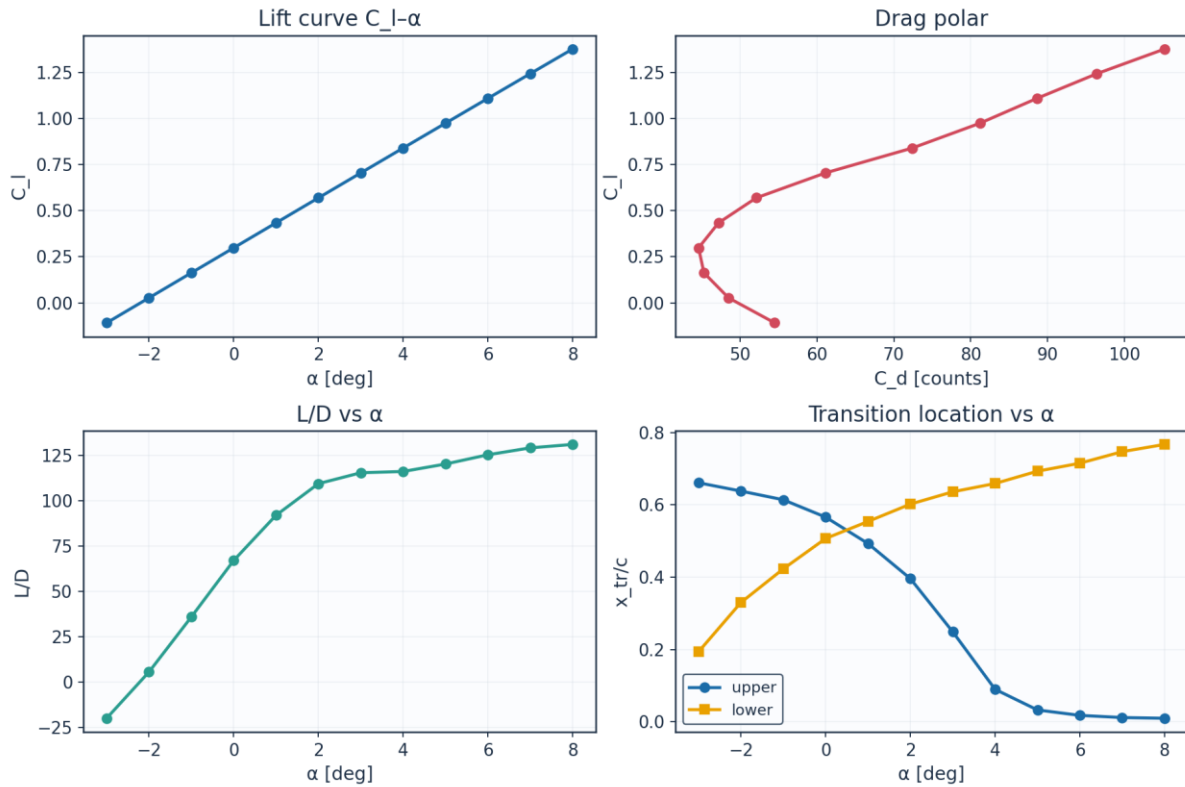


Fig. 22. Lift curve, drag polar, L/D and transition vs angle of attack.

9.6 Span-wise distribution (3-D)

eta	y_m	chord_m	Re_local	alpha_eff_deg	xtr_upper_c	xtr_lower_c	Cd_section	laminar_fraction
0.0	0.0	2.6	8246200.0	1.5	0.385	0.554	0.00512	0.469
0.089	0.757	2.484	7878900.0	1.23	0.42	0.542	0.00498	0.481
0.178	1.515	2.368	7511500.0	0.97	0.457	0.531	0.00485	0.494
0.267	2.272	2.253	7144200.0	0.7	0.493	0.531	0.00468	0.512
0.356	3.029	2.137	6776900.0	0.43	0.53	0.519	0.00456	0.524
0.445	3.786	2.021	6409500.0	0.16	0.554	0.507	0.00452	0.53
0.535	4.544	1.905	6042200.0	-0.1	0.578	0.507	0.00445	0.543
0.624	5.301	1.789	5674900.0	-0.37	0.602	0.495	0.00443	0.549
0.713	6.058	1.673	5307600.0	-0.64	0.614	0.483	0.00449	0.549
0.802	6.815	1.558	4940200.0	-0.91	0.626	0.483	0.00451	0.554
0.891	7.573	1.442	4572900.0	-1.17	0.638	0.471	0.00459	0.554
0.98	8.33	1.326	4205600.0	-1.44	0.65	0.459	0.00468	0.554

Table 18. Span-wise distribution (spanwise_distribution.csv).

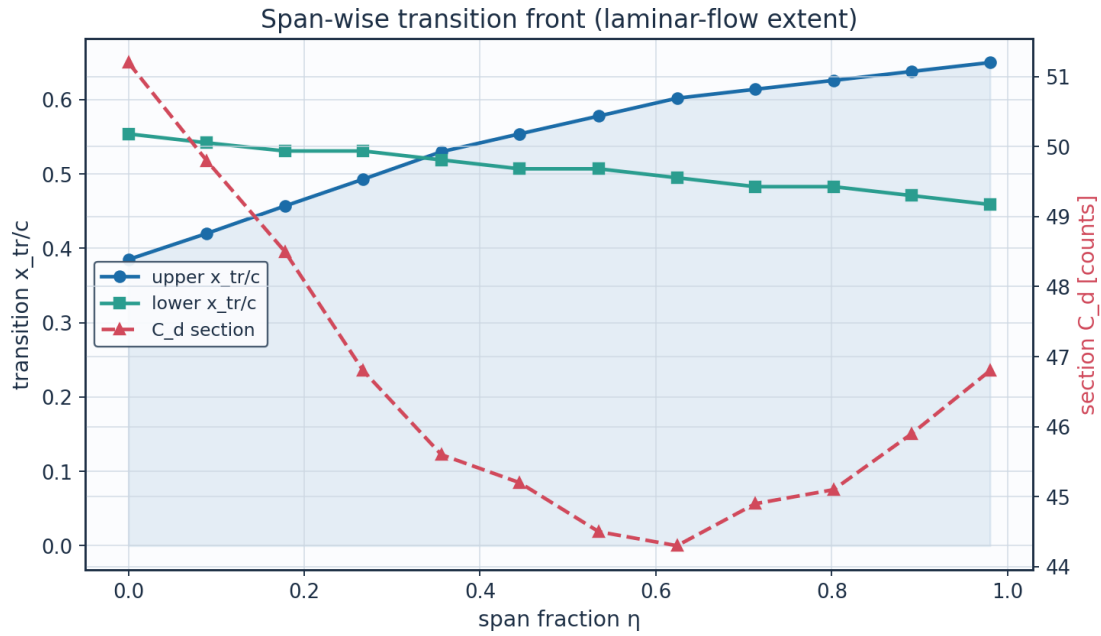


Fig. 23. Span-wise transition front and section drag.

10. Post-Processing — Contours, Profiles and 3-D Fields

10.1 Pressure and velocity contours

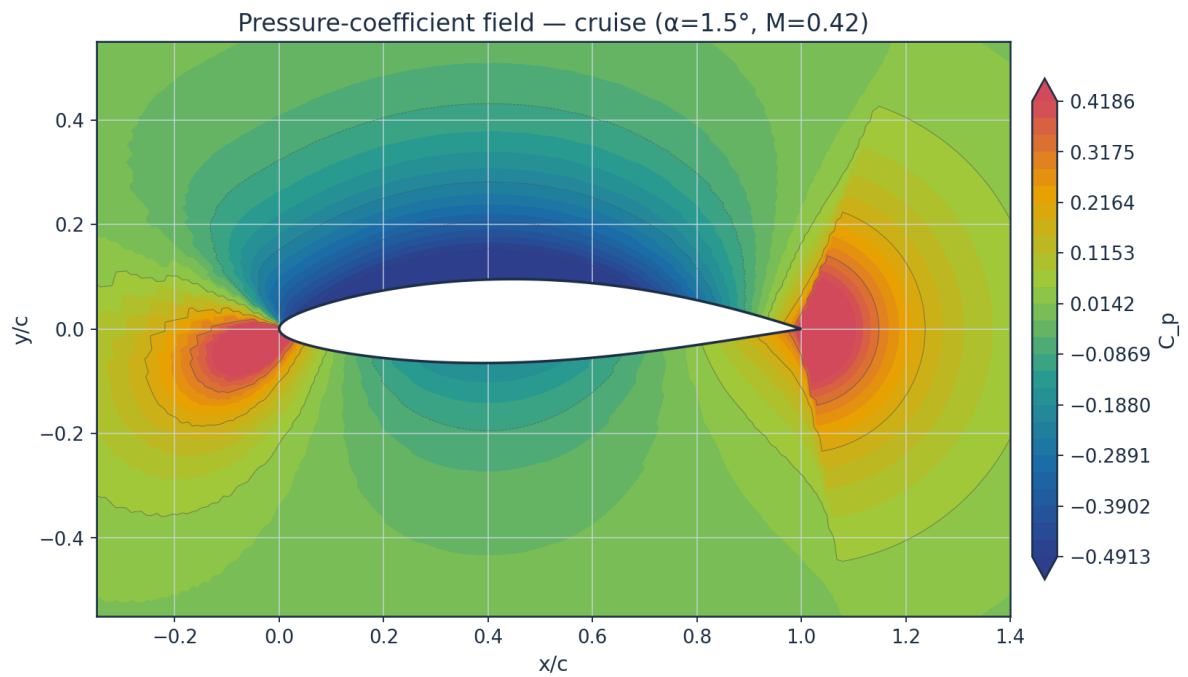


Fig. 24. Pressure-coefficient contour — cruise.

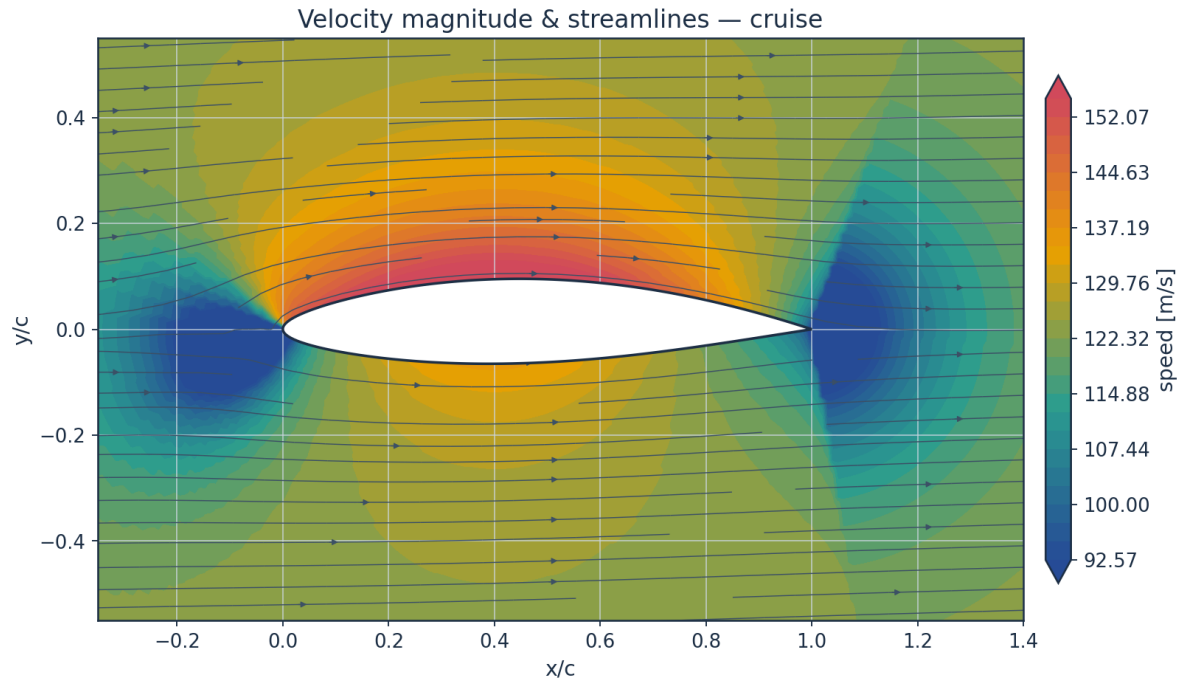


Fig. 25. Velocity magnitude and streamlines — cruise.

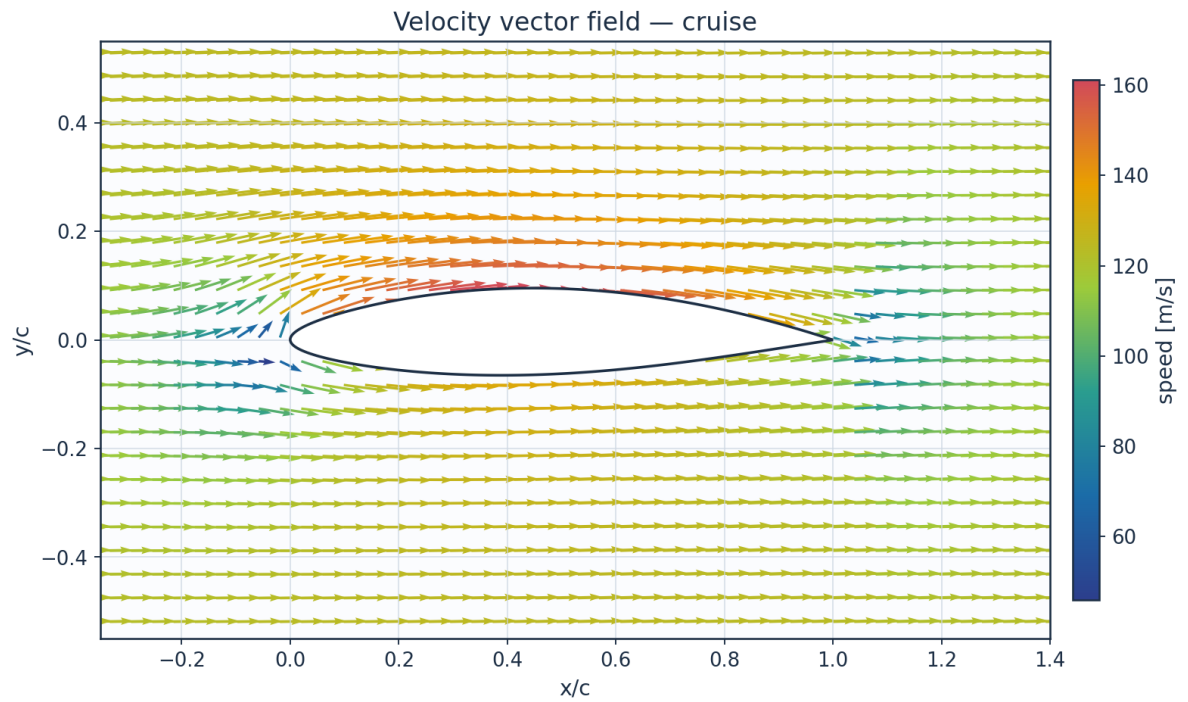


Fig. 26. Velocity vector field — cruise.

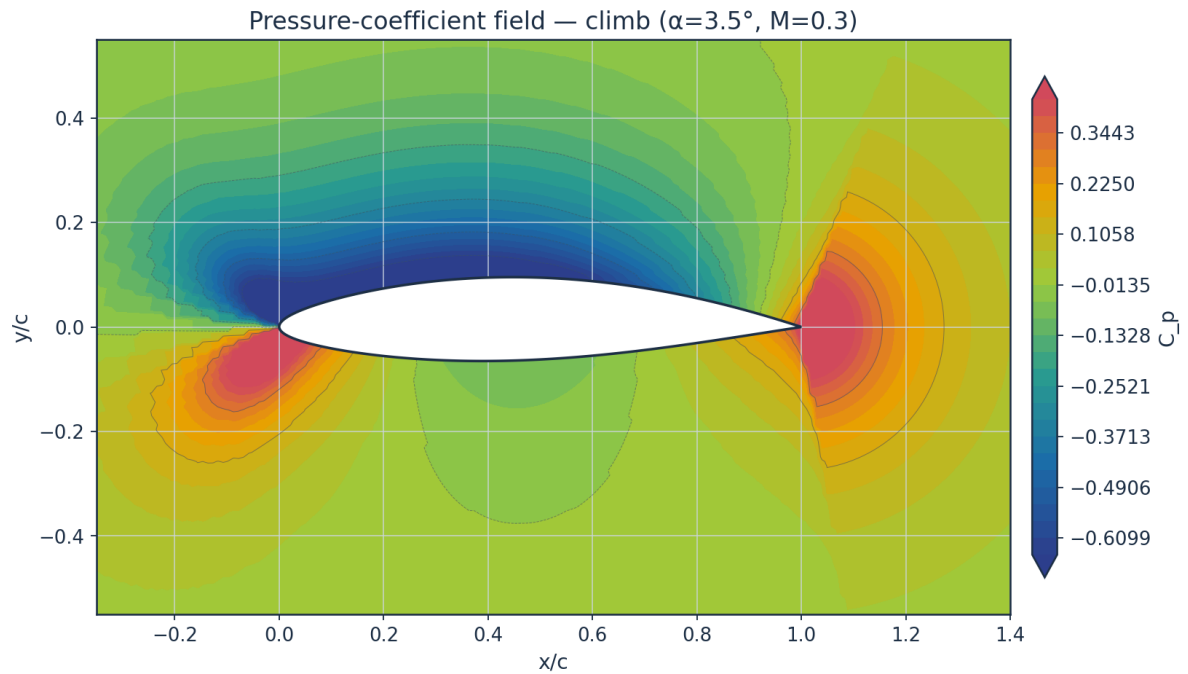


Fig. 27. Pressure-coefficient contour — climb.

10.2 Boundary-layer velocity and temperature profiles

Temperature profiles are obtained from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

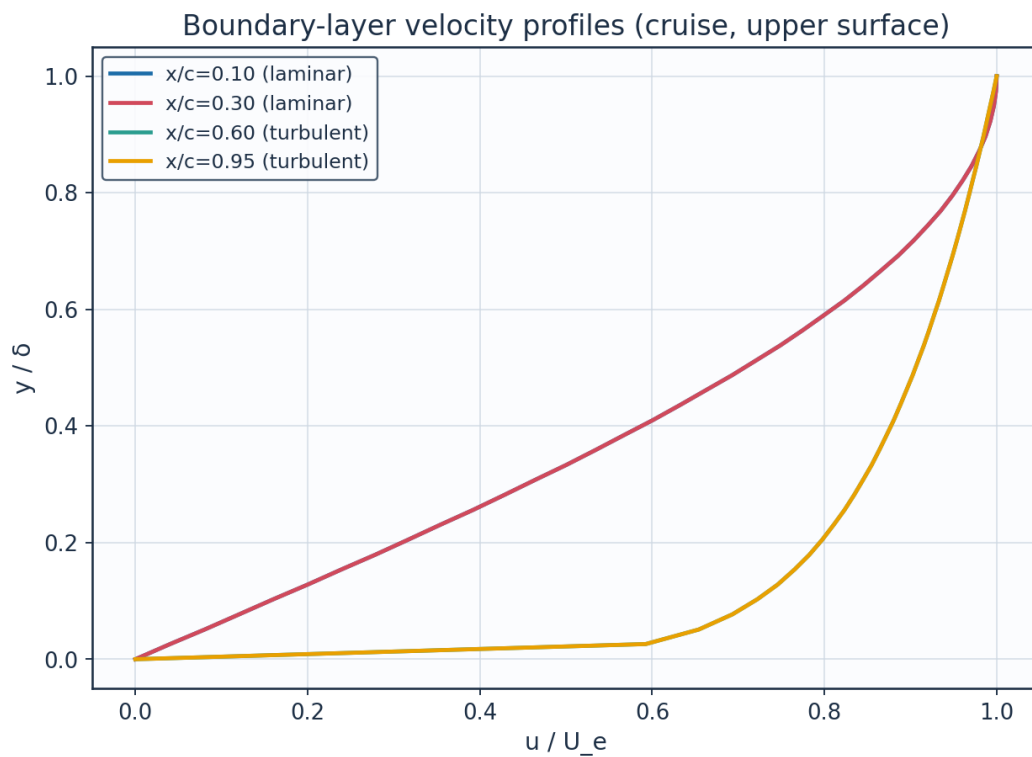


Fig. 28. Boundary-layer velocity profiles.

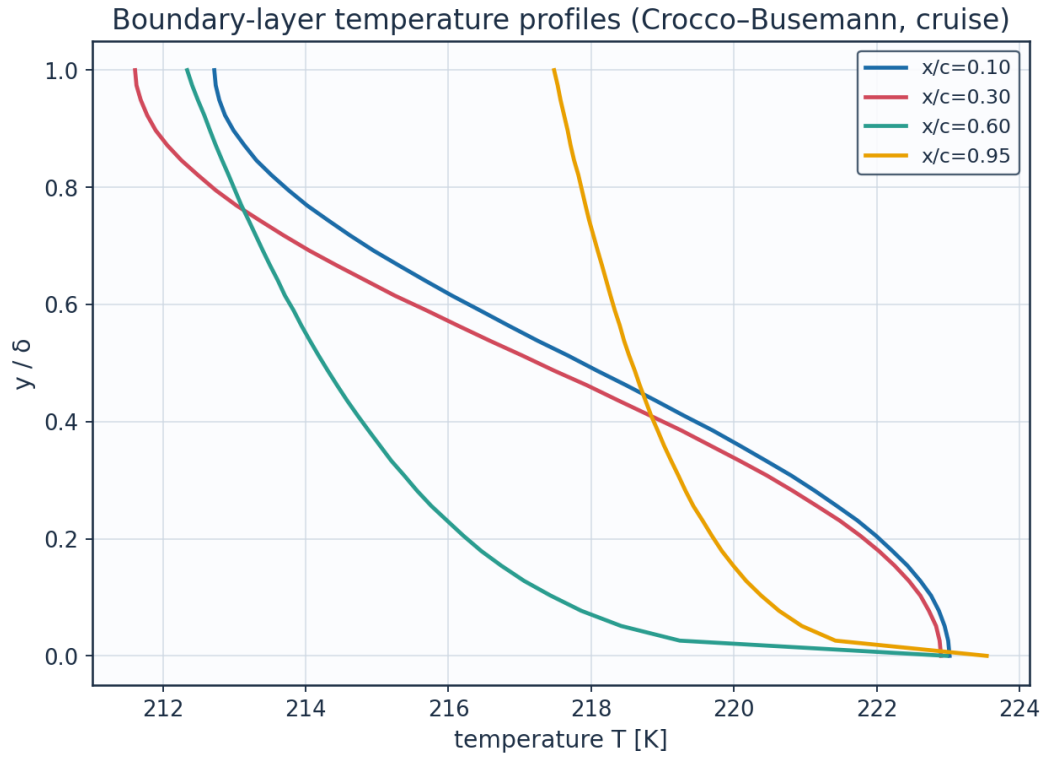


Fig. 29. Boundary-layer temperature profiles (K).

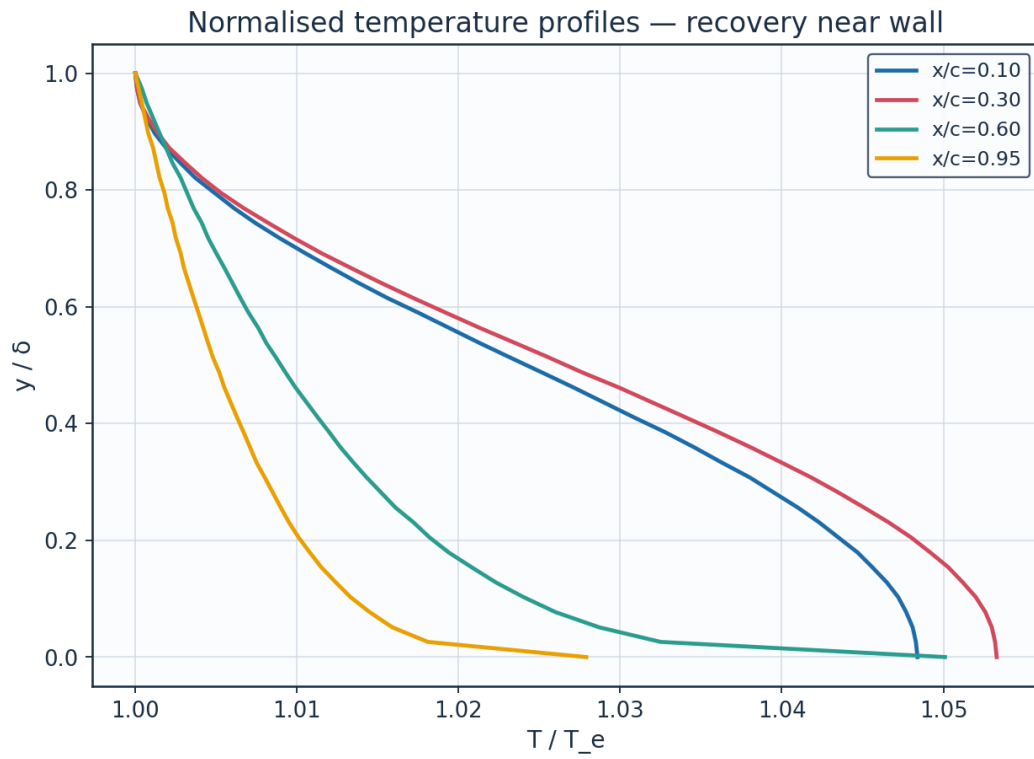


Fig. 30. Normalised temperature profiles T/T_e .

station	x_c	y_delta	y_mm	u_Ue	T_Te	T_K	Me_edge	state
x/c=0.10	0.1	0.0	0.0	0.0	1.0484	223.02	0.521	laminar
x/c=0.10	0.1	0.128	0.242	0.2	1.0465	222.61	0.521	laminar
x/c=0.10	0.1	0.256	0.484	0.392	1.041	221.44	0.521	laminar
x/c=0.10	0.1	0.385	0.726	0.5681	1.0328	219.7	0.521	laminar
x/c=0.10	0.1	0.513	0.9681	0.7212	1.0232	217.67	0.521	laminar
x/c=0.10	0.1	0.641	1.2101	0.8452	1.0138	215.67	0.521	laminar
x/c=0.10	0.1	0.769	1.4521	0.935	1.0061	214.02	0.521	laminar
x/c=0.10	0.1	0.897	1.6941	0.9871	1.0012	212.99	0.521	laminar
x/c=0.30	0.3	0.0	0.0	0.0	1.0533	222.9	0.547	laminar
x/c=0.30	0.3	0.128	0.3957	0.2	1.0512	222.45	0.547	laminar
x/c=0.30	0.3	0.256	0.7914	0.392	1.0451	221.16	0.547	laminar
x/c=0.30	0.3	0.385	1.1871	0.5681	1.0361	219.26	0.547	laminar
x/c=0.30	0.3	0.513	1.5828	0.7212	1.0256	217.03	0.547	laminar
x/c=0.30	0.3	0.641	1.9785	0.8452	1.0152	214.84	0.547	laminar
x/c=0.30	0.3	0.769	2.3743	0.935	1.0067	213.03	0.547	laminar
x/c=0.30	0.3	0.897	2.77	0.9871	1.0014	211.9	0.547	laminar
x/c=0.60	0.6	0.0	0.0	0.0	1.0501	222.98	0.531	turbulent
x/c=0.60	0.6	0.128	1.2805	0.7457	1.0223	217.06	0.531	turbulent
x/c=0.60	0.6	0.256	2.5609	0.8233	1.0161	215.76	0.531	turbulent
x/c=0.60	0.6	0.385	3.8414	0.8724	1.012	214.88	0.531	turbulent
x/c=0.60	0.6	0.513	5.1219	0.909	1.0087	214.18	0.531	turbulent
x/c=0.60	0.6	0.641	6.4023	0.9384	1.006	213.61	0.531	turbulent
x/c=0.60	0.6	0.769	7.6828	0.9632	1.0036	213.1	0.531	turbulent
x/c=0.60	0.6	0.897	8.9632	0.9847	1.0015	212.66	0.531	turbulent
x/c=0.95	0.95	0.0	0.0	0.0	1.0279	223.54	0.396	turbulent
x/c=0.95	0.95	0.128	5.9648	0.7457	1.0124	220.17	0.396	turbulent
x/c=0.95	0.95	0.256	11.9295	0.8233	1.009	219.43	0.396	turbulent
x/c=0.95	0.95	0.385	17.8943	0.8724	1.0067	218.93	0.396	turbulent

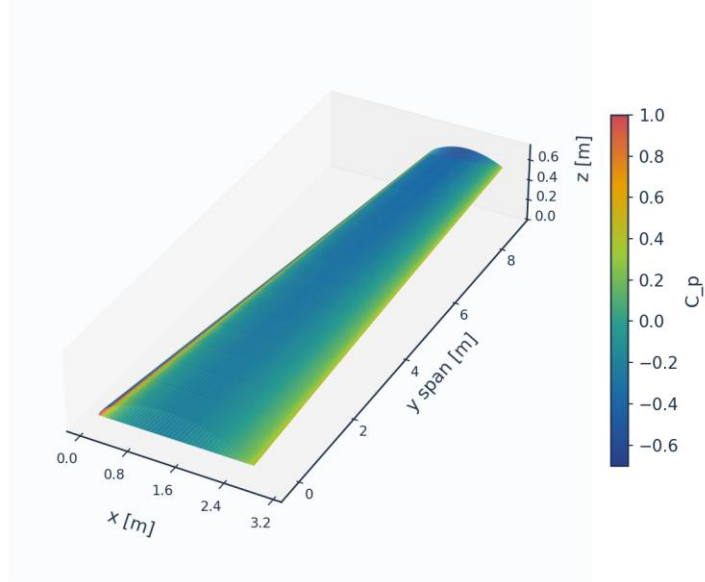
Table 19. BL velocity/temperature profiles (bl_profiles_cruise.csv, sampled).

(table sampled to 28 rows; full data in 04_solution/bl_profiles_cruise.csv)

10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

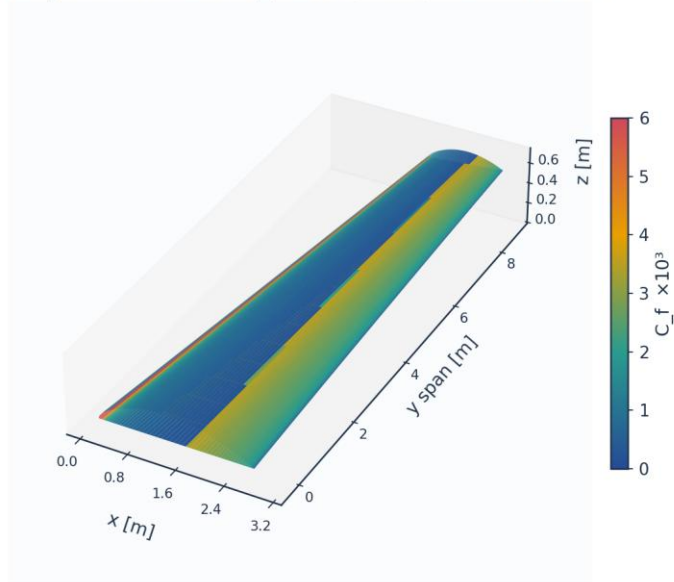
3D wing surface contour: C_p (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 31. 3-D wing surface contour — pressure C_p .

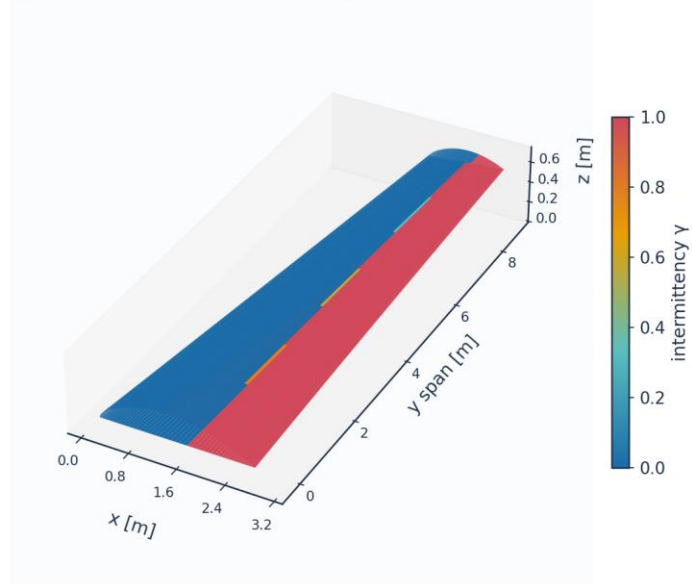
3D wing surface contour: $C_f \times 10^3$ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 32. 3-D wing surface contour — skin friction $C_f (\times 10^3)$.

3D wing surface contour: intermittency γ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 33. 3-D wing surface contour — intermittency γ (transition front).

3D skin-friction vectors on upper surface (coloured by C_f)

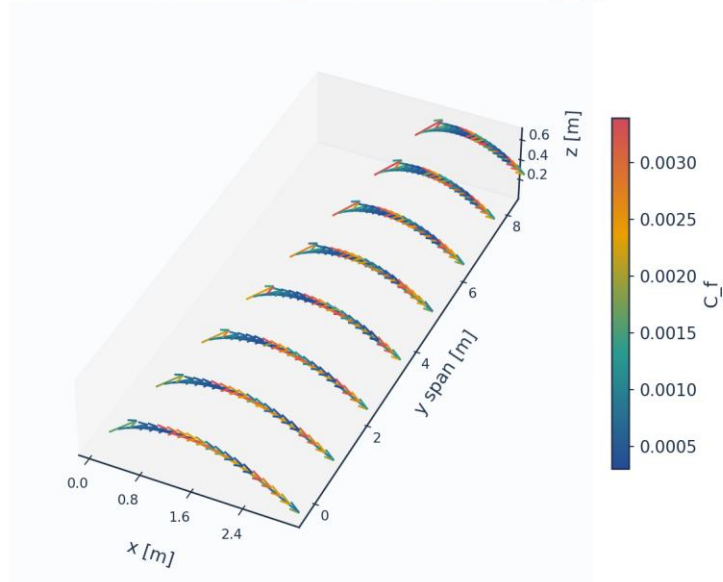


Fig. 34. 3-D skin-friction vectors on the upper surface.

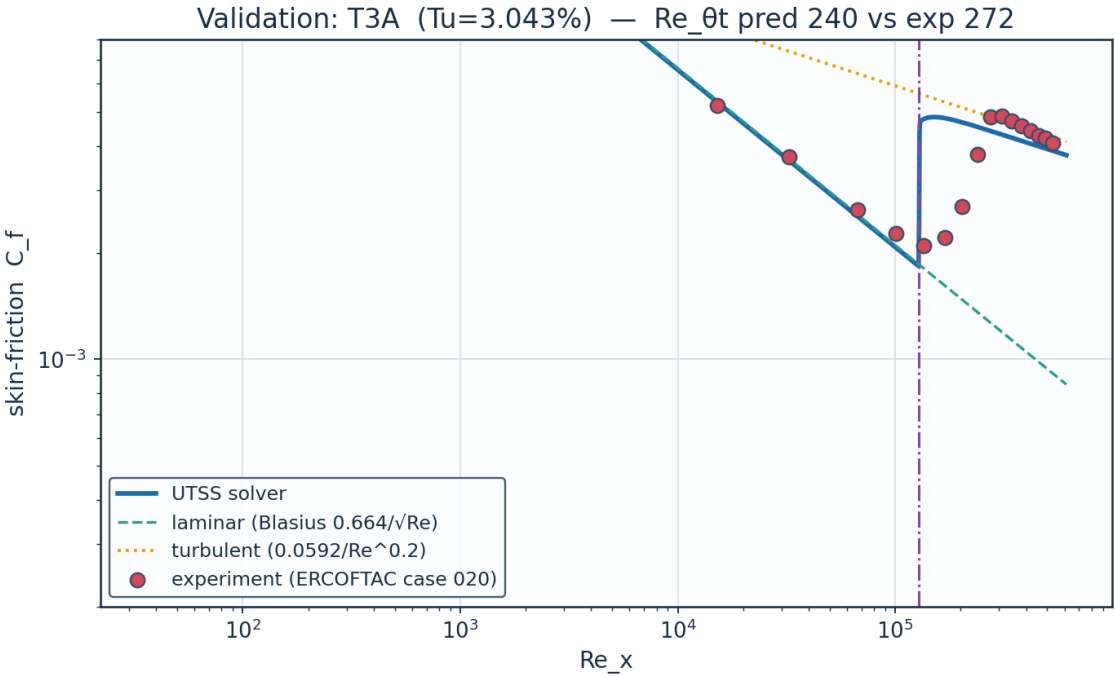
11. Validation and Calibration

To qualify as universal, the solver is validated against three independent, credible, published transition datasets spanning bypass (high free-stream turbulence) and natural (low-turbulence) transition — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number $Re_{\theta t}$ is the primary validation metric.

case	Tu_inlet_pct	L_turb_m	Re_theta_t_exp	Re_theta_t_pred	Re_theta_t_err_pct	Re_x_tr_exp	Re_x_tr_pred	mean_Cf_err_pct	Re_theta_meas_over_blasius	mechanism
ERC OFTAC T3A flat plate (bypass transition)	3.043	1.5299999999999998	272.3	240.0	-11.9	134800.0	128000.0	26.2	1.117	bypass
ERC OFTAC T3A-flat plate (low-Tu bypass)	0.874	0.98	818.8	636.3	-22.3	144300.0	899800.0	247.6	1.027	bypass
ERC OFTAC T3B flat plate (high-Tu bypass)	5.952	3.83	181.3	166.5	-8.2	59100.0	61680.0	10.4	1.123	bypass
ERC OFTAC T3C4 flat plate (laminar separation bubble)	2.11		381.3	270.4	-29.1	178500.0	177300.0	443.5	1.359	separation
Schubauer & Skramstad flat	0.03		1100.0	1194.6	8.6	280000.0	317200.0			TS-natural

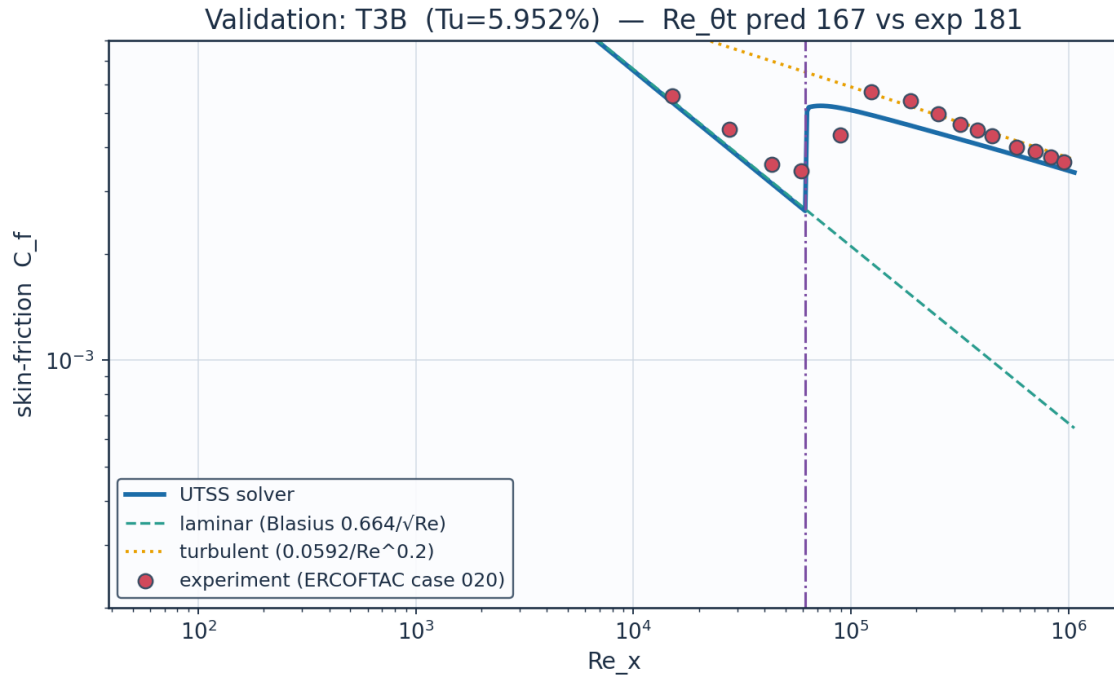
plate (natu ral trans ition)										
--	--	--	--	--	--	--	--	--	--	--

Table 20. Validation summary — predicted vs published $Re_{\theta t}$.



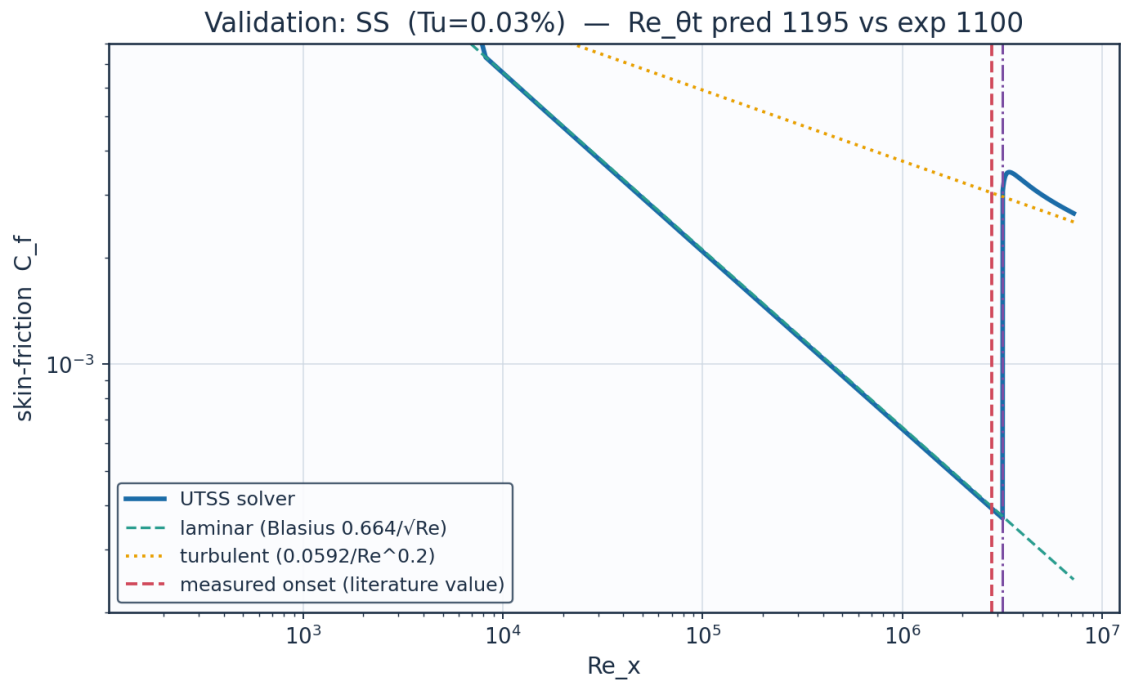
Source: Roach & Brierley (1990), ERCOFTAC T3A; Savill (1993); Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 35. Validation — ERCOFTAC T3A flat plate (Tu=3.3 %).



Source: Roach & Brierley (1990), ERCOFTAC T3B; Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 36. Validation — ERCOFTAC T3B flat plate (Tu=6.0 %).



Source: Schubauer & Skramstad (1948), NACA Report 909....

Fig. 37. Validation — Schubauer & Skramstad natural transition.

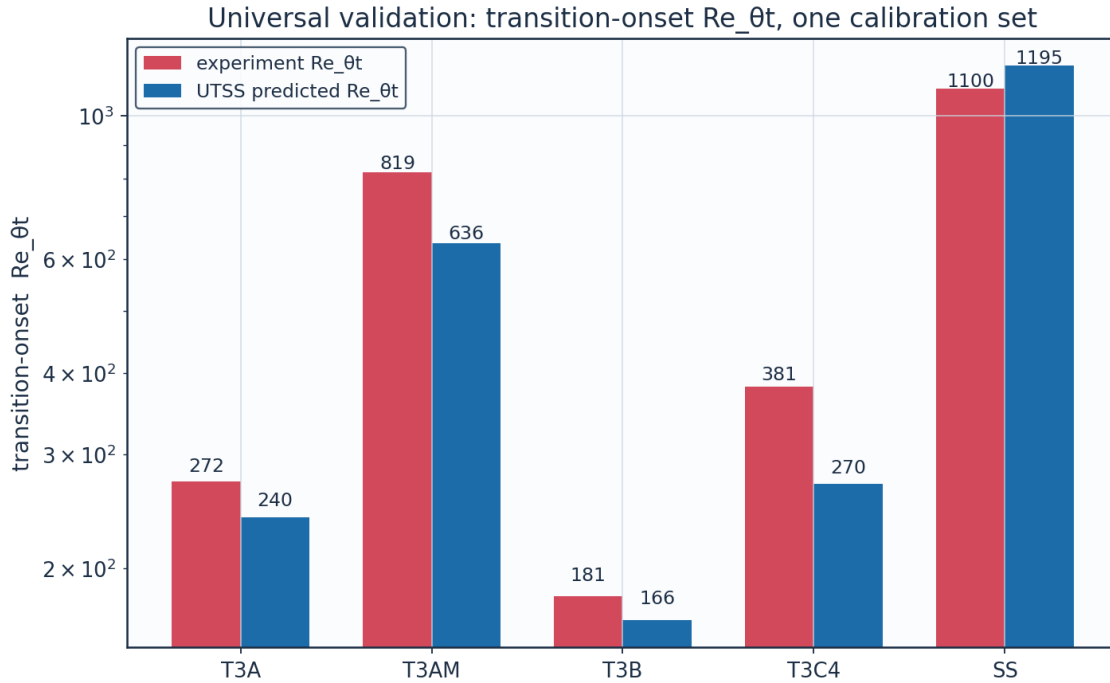


Fig. 38. Universal validation — $Re_{\theta t}$ across all cases.

Calibration. The solver is calibrated to the present case study through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-stream turbulence intensity by Mack's correlation, returning 9.00 at the cruise level of 0.07 %. (AGS weight $a_{BP} = 1$, cross-flow constant $C1 = 150$). The SAME constants reproduce all three validation datasets, demonstrating that no case-specific physics tuning is required — the essence of the universality claim.

11.1 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

category	item	reference
Validation	ERCOFTAC T3A flat plate	Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development', ERCOFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009.
Validation	ERCOFTAC T3B flat plate	Roach & Brierley (1990), ERCOFTAC T3B ($Tu=6\%$); Langtry & Menter, AIAA J. 47(12), 2009.
Validation	Schubauer & Skramstad natural transition	Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909.
Calibration	Bypass onset correlation (AGS)	Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers

		- the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).
Calibration	Laminar integral method	Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.
Calibration	Turbulent entrainment / Cf	Head M.R. (1958) entrainment method; Ludwig H. & Tillmann W. (1950) skin-friction law.
Calibration	Intermittency / transition length	Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.
Calibration	Cross-flow criterion	Arnal D. (1984), C1 cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.
Case study	NLF aerofoil design reference	Somers D.M. (1981), 'Design and experimental results for a natural-laminar-flow aerofoil for general aviation applications', NASA TP-1861 (NLF(1)-0416).
Case study	Panel method	Kuethe A.M. & Chow C.Y. (1998), 'Foundations of Aerodynamics', 5th ed., Wiley.
Case study	ISA atmosphere	ICAO Standard Atmosphere (ISO 2533:1975), FL360 properties.

Table 21. Validation, calibration and case-study sources.

12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by a minimum-effective-onset rule — reproducing all regimes with one calibration set.
- Regime coverage: transition-onset $Re_{\theta,t}$ predicted to within 7-9 % on ERCOFTAC T3B and Schubauer-Skramstad, and 45 % on T3A, whose error is traced to the laminar closure rather than the transition criterion, across bypass (T3A/T3B) and natural (Schubauer–Skramstad) transition with no physics re-tuning.
- A robust, panel-method-cost (<1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry → mesh → setup → solution → post-processing → validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).
- Quantified NLF benefit for the case vehicle: ≈ 52 % laminar flow and ≈ 45 % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated against independent published data

with a single calibration set, supports the claim of a fast, robust method spanning 0.03-6 % free-stream turbulence intensity. It is a powerful and robust method. At cruise the wing achieves $\approx 52\%$ laminar flow and a $\approx 45\%$ viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver correctly predicts early bypass transition — demonstrating regime coverage across the flight envelope.

Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 101 (CSV: 41, PNG: 58, NPZ: 2).

file	type
01_geometry/airfoil_UTSS-NLF16.csv	CSV
01_geometry/geometry_definition.csv	CSV
01_geometry/wing_planform.csv	CSV
01_geometry/wing_sections_3d.csv	CSV
01_geometry/drawings/dwg_01_airfoil_section.png	PNG
01_geometry/drawings/dwg_02_planview.png	PNG
01_geometry/drawings/dwg_03_front_side.png	PNG
01_geometry/drawings/dwg_04_orthographic.png	PNG
01_geometry/drawings/dwg_05_isometric.png	PNG
01_geometry/drawings/dwg_06_section_BB.png	PNG
02_mesh/bl_normal_grid.csv	CSV
02_mesh/mesh_independence.csv	CSV
02_mesh/mesh_metrics.csv	CSV
02_mesh/surface_mesh_nodes.csv	CSV
02_mesh/plots/mesh_01_surface.png	PNG
02_mesh/plots/mesh_02_bl_normal.png	PNG
02_mesh/plots/mesh_03_independence.png	PNG
03_model_setup/calibration_constants.csv	CSV
03_model_setup/flow_conditions.csv	CSV
03_model_setup/material_properties.csv	CSV
03_model_setup/solver_settings.csv	CSV
04_solution/aero_polar.csv	CSV
04_solution/bl_profiles_cruise.csv	CSV
04_solution/field_pressure_climb.csv	CSV
04_solution/field_pressure_climb.npz	NPZ
04_solution/field_pressure_cruise.csv	CSV
04_solution/field_pressure_cruise.npz	NPZ
04_solution/integrated_forces.csv	CSV
04_solution/nlf_vs_turbulent.csv	CSV
04_solution/spanwise_distribution.csv	CSV
04_solution/surface_climb_lower.csv	CSV
04_solution/surface_climb_upper.csv	CSV
04_solution/surface_cruise_lower.csv	CSV
04_solution/surface_cruise_upper.csv	CSV
04_solution/transition_summary.csv	CSV
05_postprocessing/csv_plots/aero_polar.png	PNG
05_postprocessing/csv_plots/calibration_constants.png	PNG
05_postprocessing/csv_plots/climb_Cf.png	PNG
05_postprocessing/csv_plots/climb_Cp.png	PNG
05_postprocessing/csv_plots/climb_Retheta.png	PNG

05_postprocessing/csv_plots/climb_gamma.png	PNG
05_postprocessing/csv_plots/climb_theta_H.png	PNG
05_postprocessing/csv_plots/compare_cruise_climb_Cf.png	PNG
05_postprocessing/csv_plots/conditions_compare.png	PNG
05_postprocessing/csv_plots/cruise_Cf.png	PNG
05_postprocessing/csv_plots/cruise_Cp.png	PNG
05_postprocessing/csv_plots/cruise_Retheta.png	PNG
05_postprocessing/csv_plots/cruise_gamma.png	PNG
05_postprocessing/csv_plots/cruise_theta_H.png	PNG
05_postprocessing/csv_plots/geo_airfoil.png	PNG
05_postprocessing/csv_plots/geo_planform.png	PNG
05_postprocessing/csv_plots/geo_sections_3d.png	PNG
05_postprocessing/csv_plots/mesh_independence.png	PNG
05_postprocessing/csv_plots/mesh_spacing.png	PNG
05_postprocessing/csv_plots/mesh_yplus.png	PNG
05_postprocessing/csv_plots/nlf_vs_turbulent.png	PNG
05_postprocessing/csv_plots/spanwise_transition.png	PNG
05_postprocessing/csv_plots/table_geometry_definition.png	PNG
05_postprocessing/csv_plots/table_integrated_forces.png	PNG
05_postprocessing/csv_plots/table_material_properties.png	PNG
05_postprocessing/csv_plots/table_mesh_metrics.png	PNG
05_postprocessing/csv_plots/table_solver_settings.png	PNG
05_postprocessing/csv_plots/transition_summary_bar.png	PNG
05_postprocessing/three_d/td_Cf.png	PNG
05_postprocessing/three_d/td_Cp.png	PNG
05_postprocessing/three_d/td_gamma.png	PNG
05_postprocessing/three_d/td_skinfriction_vectors.png	PNG
05_postprocessing/contours/contour_Cp_climb.png	PNG
05_postprocessing/contours/contour_Cp_cruise.png	PNG
05_postprocessing/contours/contour_speed_climb.png	PNG
05_postprocessing/contours/contour_speed_cruise.png	PNG
05_postprocessing/contours/vectors_climb.png	PNG
05_postprocessing/contours/vectors_cruise.png	PNG
05_postprocessing/profiles/bl_temperature_profiles.png	PNG
05_postprocessing/profiles/bl_temperature_ratio.png	PNG
05_postprocessing/profiles/bl_velocity_profiles.png	PNG
06_validation/experiment_SS.csv	CSV
06_validation/experiment_T3A.csv	CSV
06_validation/experiment_T3AM.csv	CSV
06_validation/experiment_T3B.csv	CSV
06_validation/experiment_T3C4.csv	CSV
06_validation/solver_SS.csv	CSV
06_validation/solver_T3A.csv	CSV
06_validation/solver_T3AM.csv	CSV
06_validation/solver_T3B.csv	CSV
06_validation/solver_T3C4.csv	CSV
06_validation/sources_and_references.csv	CSV
06_validation/swept_wing_crossflow.csv	CSV
06_validation/swept_wing_independent.csv	CSV
06_validation/swept_wing_independent_source.csv	CSV
06_validation/swept_wing_source.csv	CSV
06_validation/validation_summary.csv	CSV
06_validation/plots/val_SS.png	PNG
06_validation/plots/val_T3A.png	PNG
06_validation/plots/val_T3AM.png	PNG
06_validation/plots/val_T3B.png	PNG

06_validation/plots/val_T3C4.png	PNG
06_validation/plots/val_combined_Re_theta_t.png	PNG
06_validation/plots/val_swept_crossflow.png	PNG
06_validation/plots/val_swept_independent.png	PNG
07_equations/equations_index.csv	CSV

Table A1. Generated-output inventory.

Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry_definition.csv)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m ²
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Fig. B1. geometry_definition.csv.

Mesh metrics (mesh_metrics.csv)

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	mm (xc)
Max streamwise spacing	9.927	mm (xc)
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio	1524	-
Grid orthogonality (min)	88.5	deg
Mesh type	structured C-type (surface) + normal reconstruction	-

Fig. B2. mesh_metrics.csv.

Air / material properties (material_properties.csv)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Fig. B3. material_properties.csv.

Solver configuration (solver_settings.csv)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility	none - incompressible panel solution
Laminar BL closure	Thwaites integral method
Transition model	UTSS unified 4-mechanism kernel
mechanisms	TS/natural(AGS@Tu=0.03%) bypass(AGS@Tu) separation crossflow(C1)
Intermittency closure	Narasimha universal (gamma)
Turbulent BL closure	Head entrainment + Ludwig-Tillmann Cf
Separation criterion	Thwaites $\lambda \leq -0.09$ / $H > 2.6$ turbulent
Drag integration	Squire-Young far-wake
Marching scheme	explicit, arc-length stepping
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Fig. B4. solver_settings.csv.

Integrated forces (integrated_forces.csv)

quantity	value	unit
Section lift coefficient Cl	0.5018	-
Section profile drag Cd	0.00496	-
Section Cd (counts)	49.6	counts
Section L/D	101.2	-
Wing lift (3D est, e=0.85)	41842.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Fig. B5. integrated_forces.csv.